



Epigenetics: DNA methylation and epigenetic control

Methylation-Based Breast Cancer Screening: primary tissue genomic DNA and circulating cell-free DNA (ccfDNA)

April 12 2025

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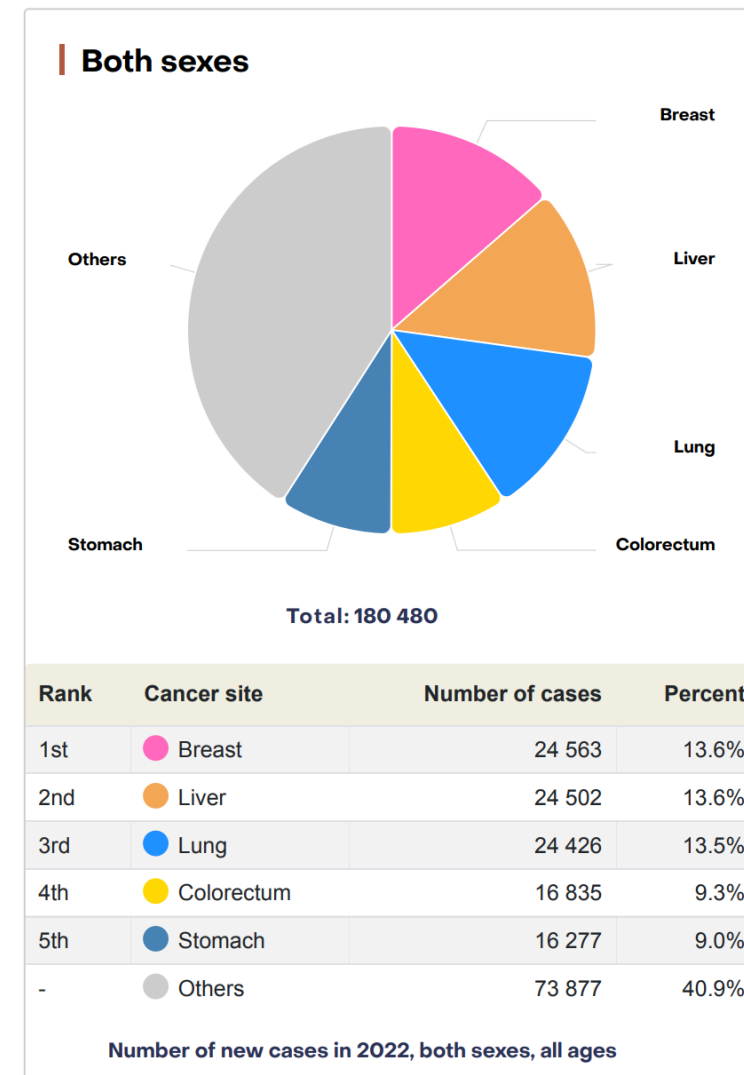
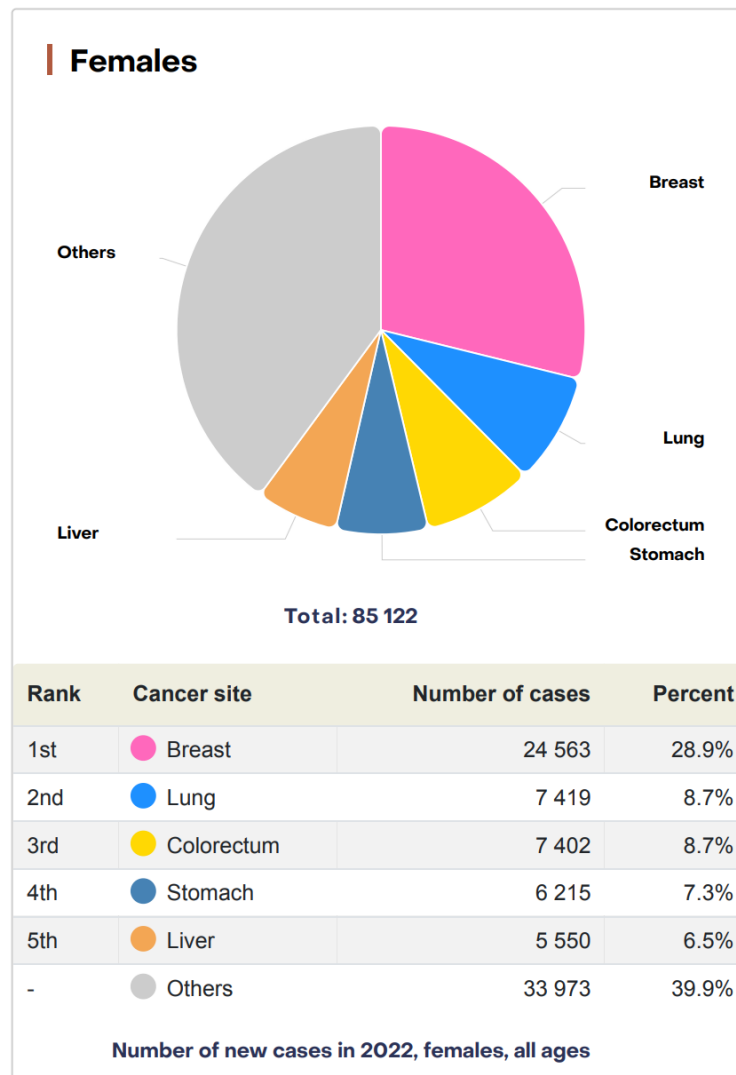
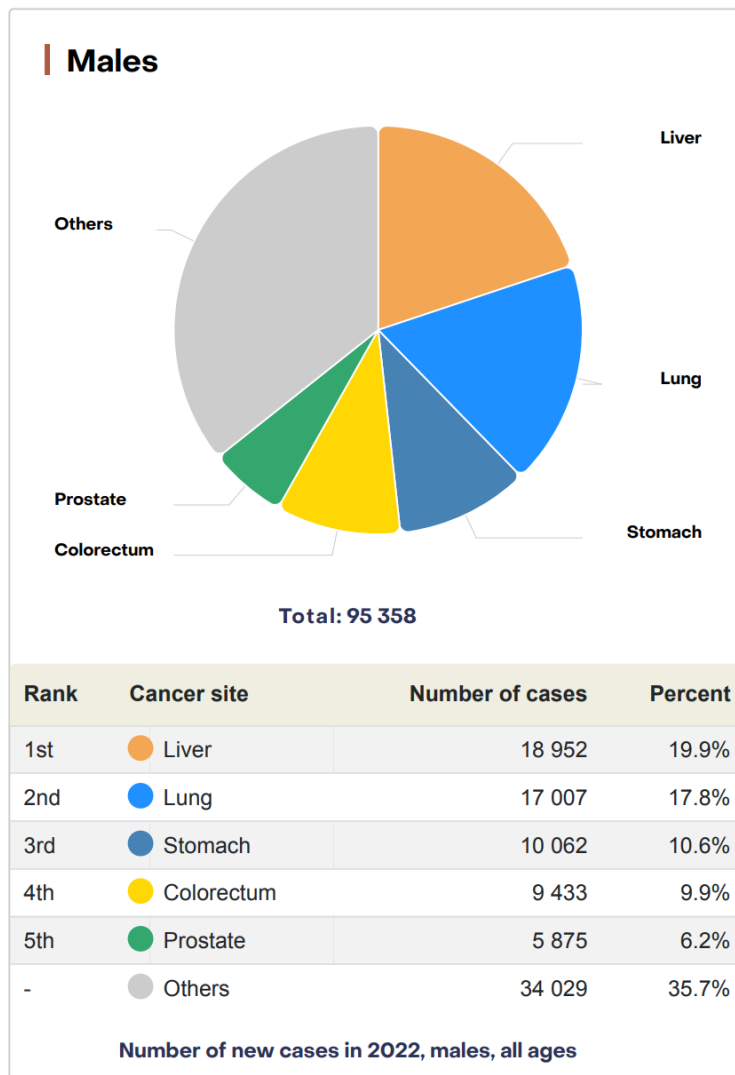
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Content

- Breast cancer in Vietnam
- Screening information for breast cancer
- Diagnosing breast cancer
- Epigenetics: DNA methylation and epigenetic control
- Breast Cancer DNA methylation signatures in circulating tumour DNA

Breast cancer is ranked the first top number of cases in Vietnam

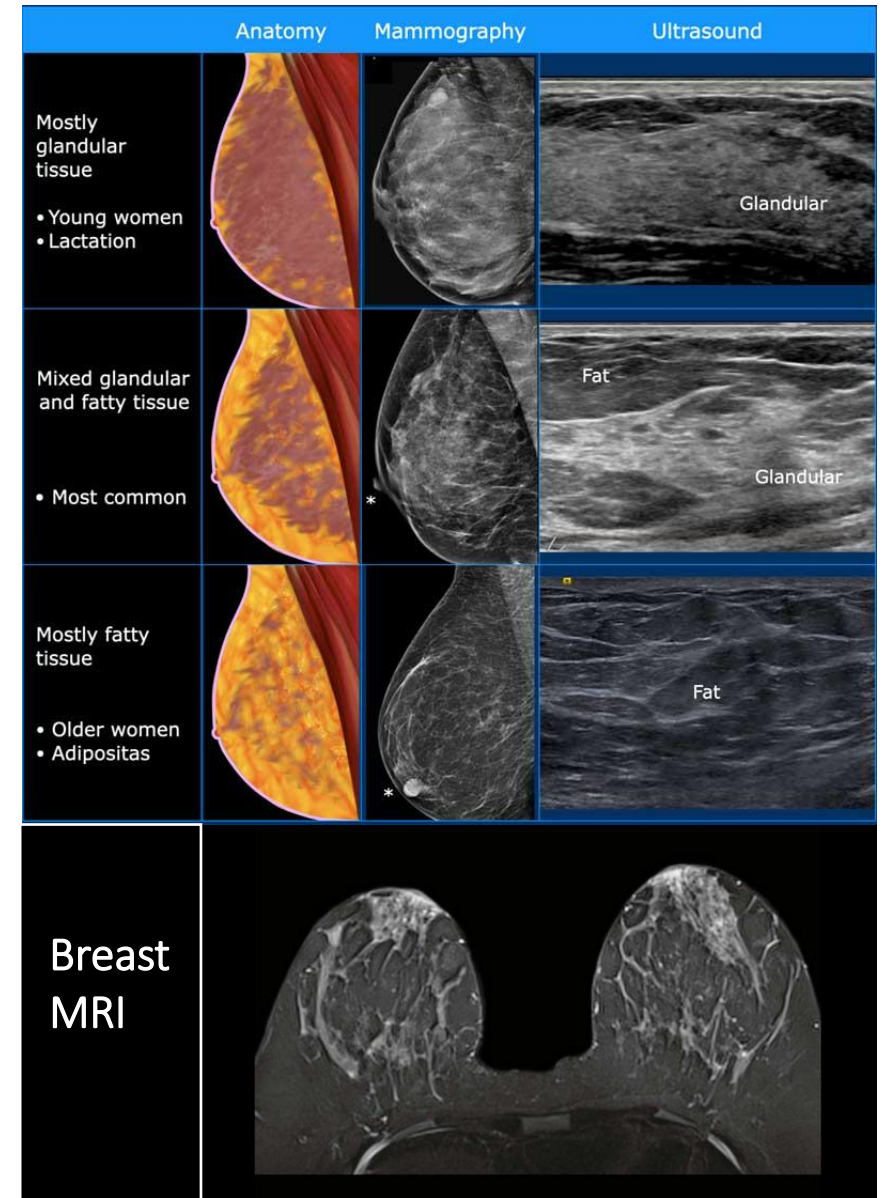
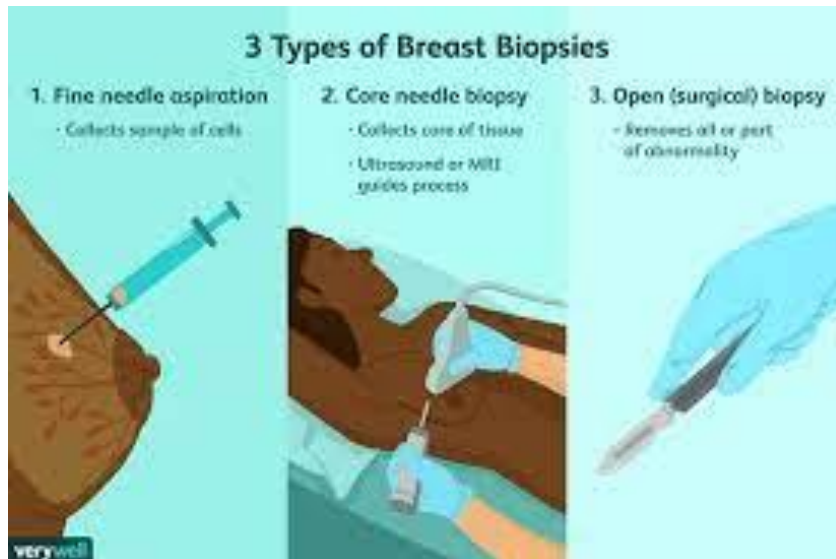


Screening information for breast cancer: goals

- Screening is used to look for cancer before you have any symptoms or signs.
- The overall goals of breast cancer screening are to:
 - ❑ Detect the cancer at an earlier stage of disease, which is when the cancer is **most curable**
 - ❑ Lower the number of people who die from the disease, or **eliminate deaths** from cancer altogether
 - ❑ Identify people with a **higher risk** of developing a specific type of cancer who may need screening more often or a different type of screening due to **genetic mutations** or diseases

Screening information for breast cancer

- **Diagnostic mammogram:** 10% to 15% of the time false-negative
- **Breast ultrasound**
- **Breast magnetic resonance imaging (MRI)**
- **Biopsy** (fine-needle aspiration, core biopsy, or open biopsy)



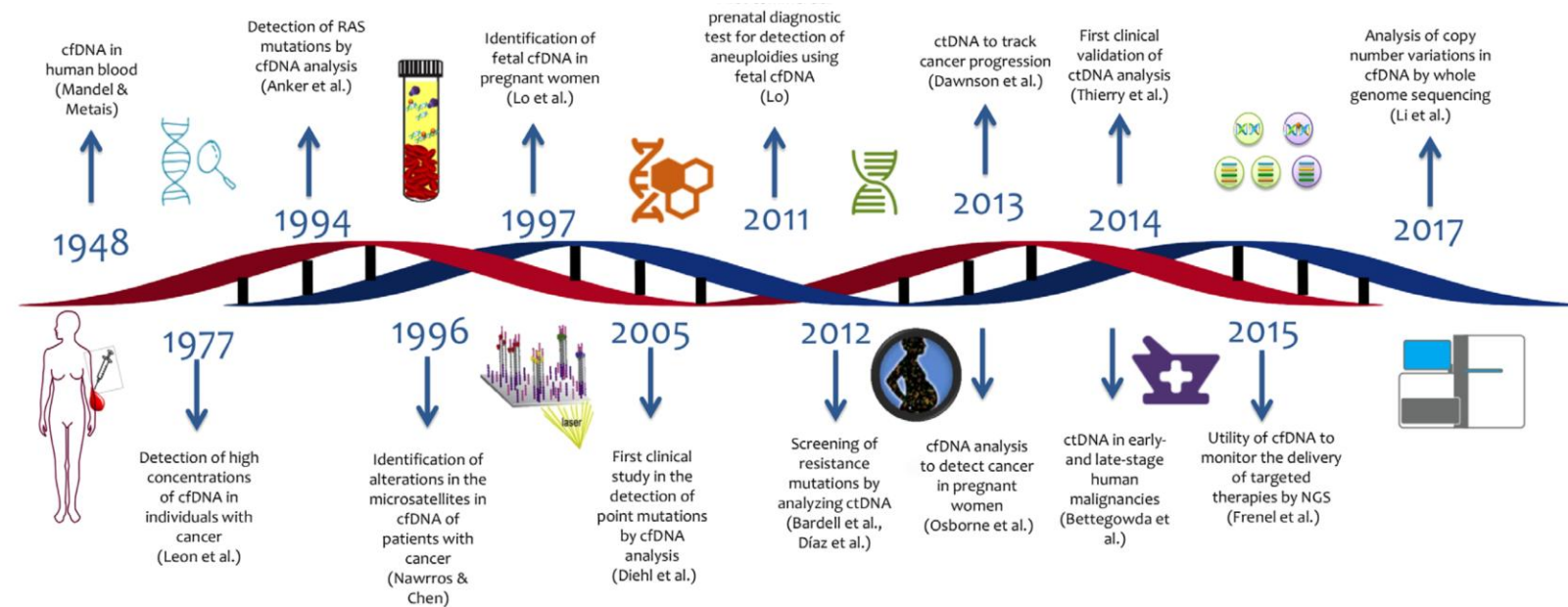
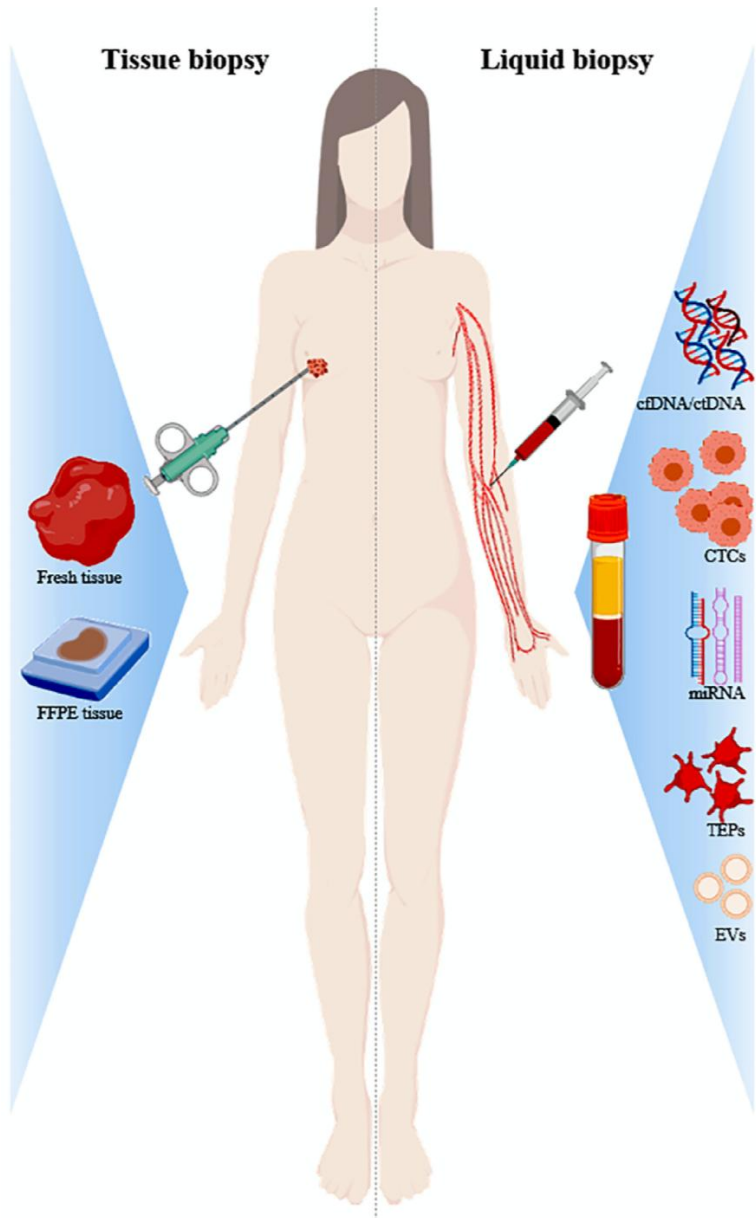
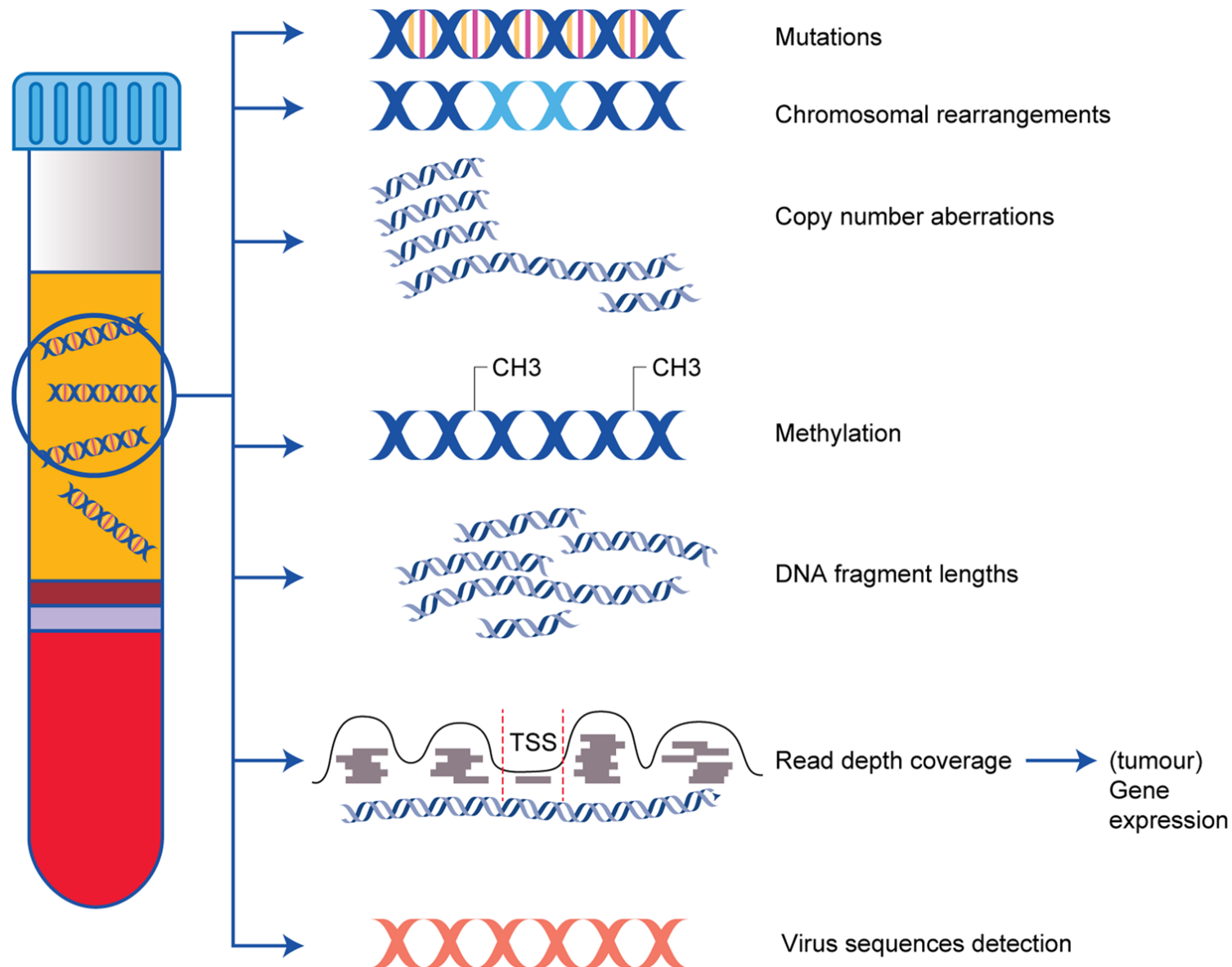


Figure 1: Timeline of liquid biopsy development.

10.18632/oncotarget.23131

Different features of ctDNA and potential clinical implications



The Human Genome

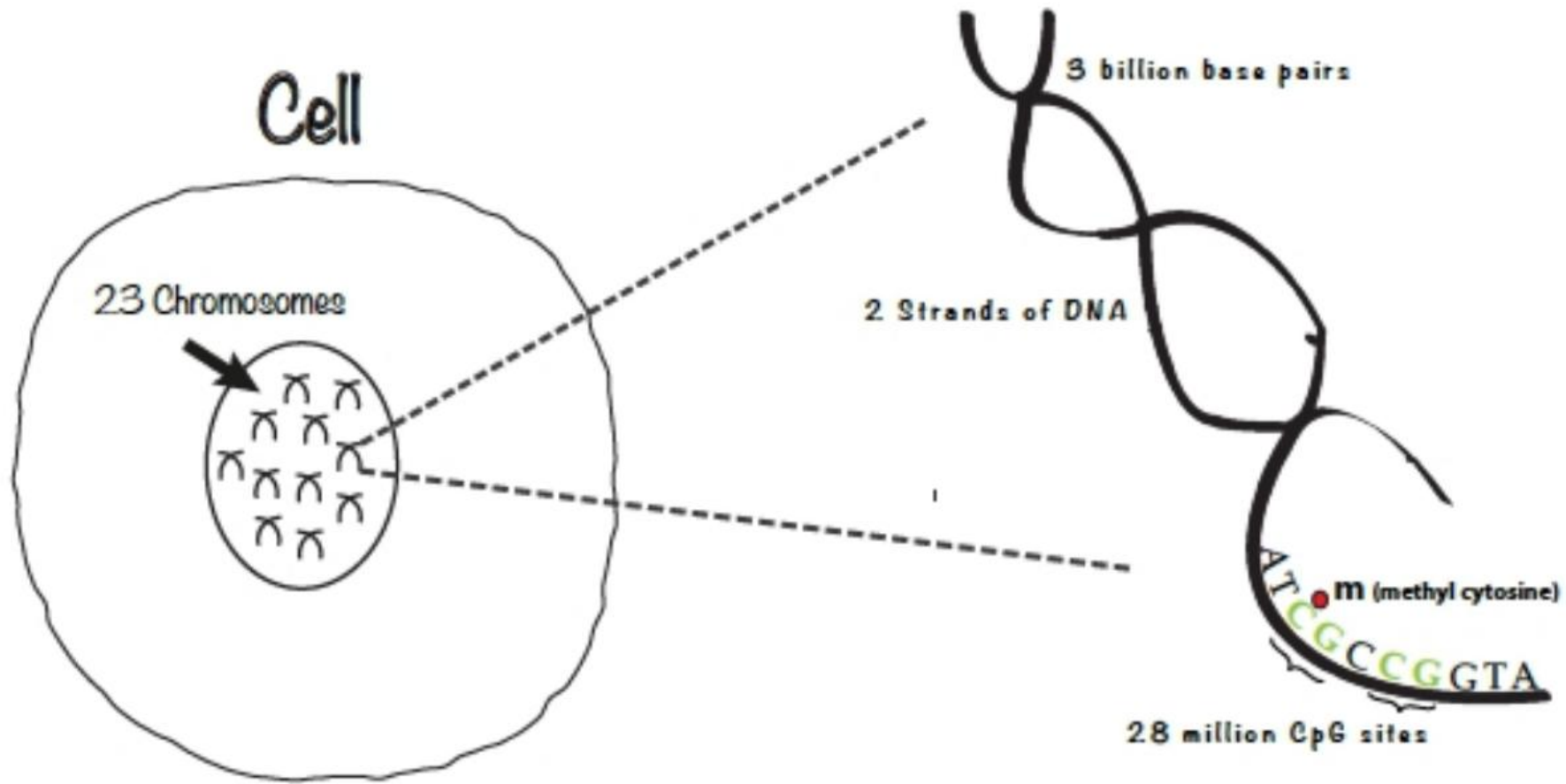
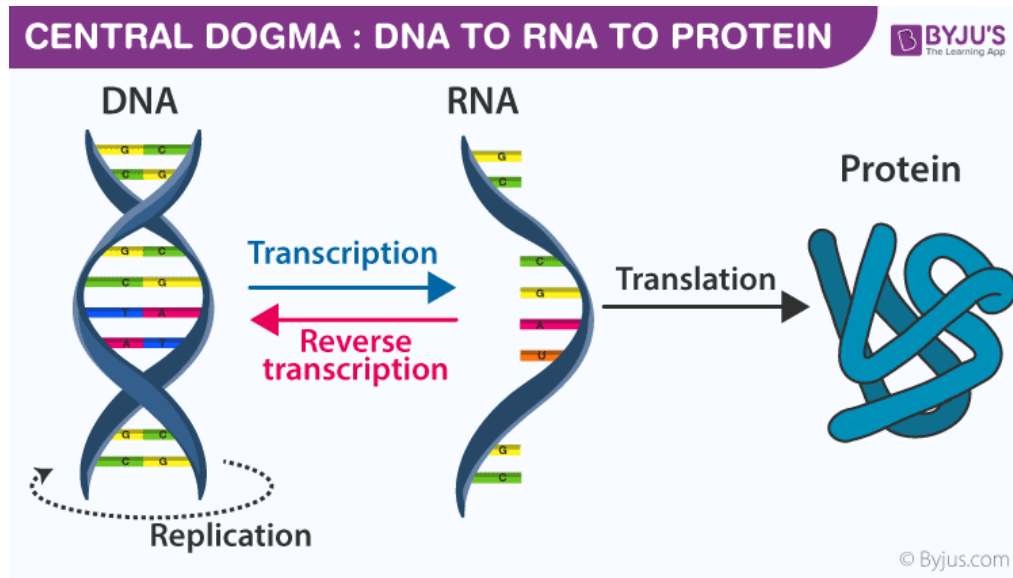


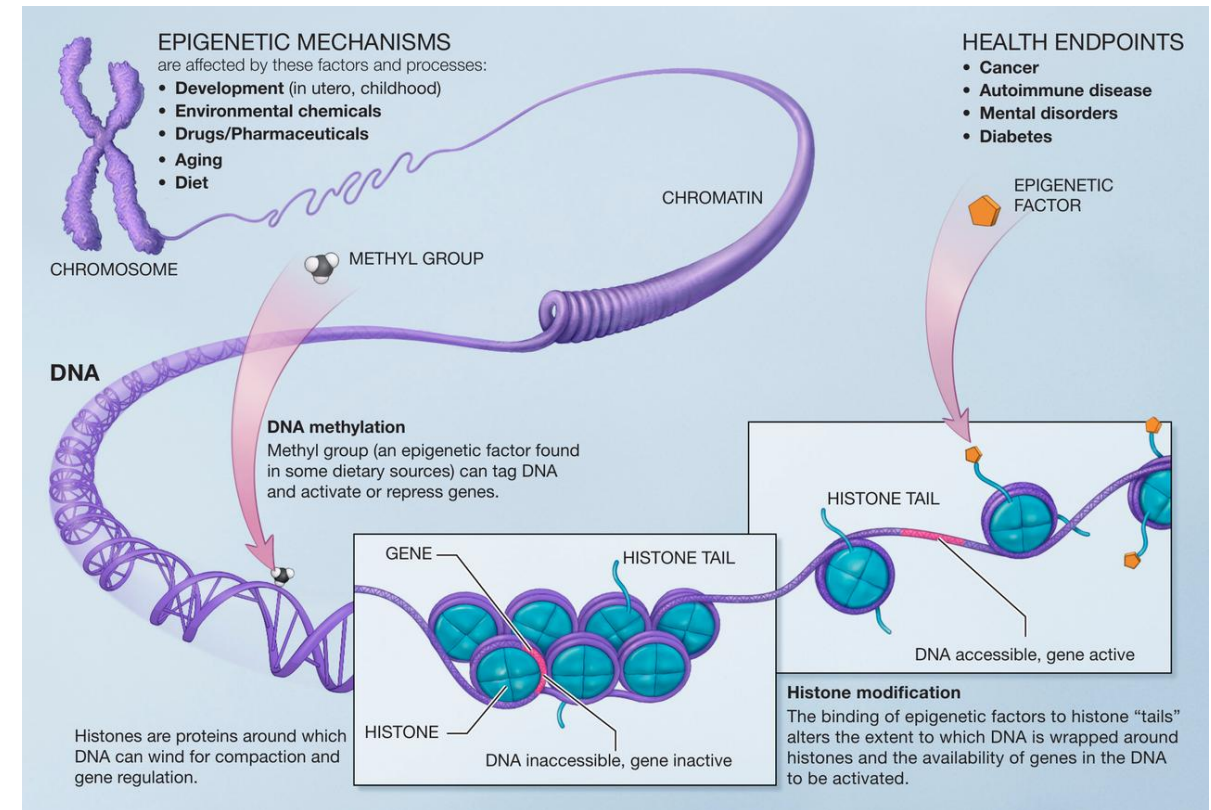
Figure 1. DNA is packaged inside cells in discrete units called chromosomes. Each cell contains 3 billion DNA base pairs, of which ~28 million are CpG sites that can be methylated.

What is Epigenetics?

Your genes play an important role in your health

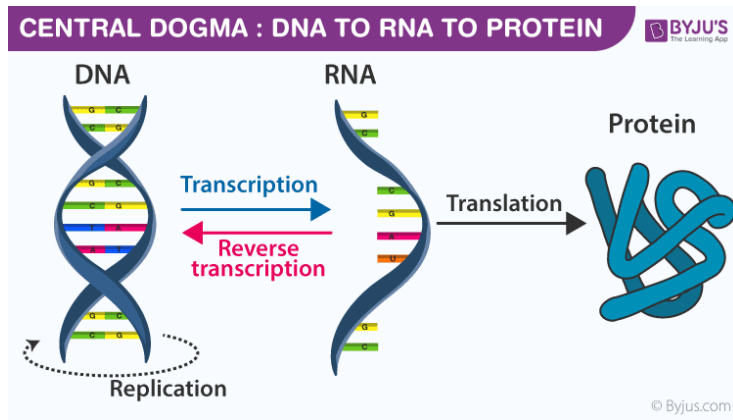


Your behaviors and environment can cause changes that affect the way your genes work



Genetics

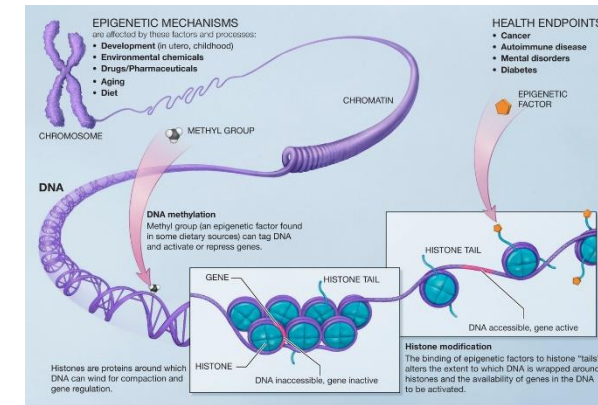
Your genes play an important role in your health



A woman without her man is nothing

Epigenetics

Your behaviors and environment can cause changes that affect the way your genes work



A woman, without her man, is nothing
A woman: without her, man is nothing

“Reading the book of life”

EPIGENETIC REGULATION OF NORMAL CELLULAR PROCESSES IS TYPICALLY DRIVEN IN A CELL TYPE-DEPENDENT MANNER. THIS REQUIRES AN EXQUISITE LEVEL OF INTERPLAY BETWEEN EPIGENETIC LAYERS INCLUDING DNA METHYLATION, NUCLEOSOME POSITIONS AND HISTONE MODIFICATIONS AMONGST OTHERS. TOGETHER, EPIGENETIC MECHANISMS ESTABLISH THE CORRECT GENE EXPRESSION PATTERNS AND HIGHER-ORDER CHROMATIN STRUCTURE, THUS THE IDENTITY OF EACH CELL. OF THESE, DNA METHYLATION IS THE BEST STUDIED EPIGENETIC MODIFICATION. PRECISE DNA METHYLATION PATTERNS ARE ESTABLISHED DURING EMBRYONIC DEVELOPMENT AND ARE MITOTICALLY HERITABLE THROUGH MULTIPLE CELLULAR DIVISIONS.

“Translating the book of life”

Epigenetic regulation of normal cellular processes is typically driven in a cell type-dependent manner. This requires an exquisite level of interplay between epigenetic layers, including DNA methylation, nucleosome positions and histone modifications, amongst others.

Together, epigenetic mechanisms establish the correct gene-expression patterns and higher-order chromatin structures; thus, the identity of each cell. Of these, DNA methylation is the best-studied epigenetic modification. Precise DNA methylation patterns are established during embryonic development and are mitotically heritable through multiple cellular divisions.

Genetics



The structure of DNA was discovered in 1953

```
GTAAGGCAGTTCGATATAGGTT
CACCGGTCTTAGGCACGTTGAG
CTGACCAGTGCTAAACGTCCAA
TGGCATTGCAGCTGGGGGTACAC
AGTCTTGTGTCACAGTCAGGTC
TATCTGTACTAGTCTTACGTAGT
CGGTACCCCATGACATCGACTT
GTGTCGGATCGATCATC
```

DNA code was mapped in 2001
– 3 billion letters (GATC)

Genetics



How does each cell know
what it is supposed to be?

```
GTAAGGCAGTTCGATATAGGTT
CACCGGTCTTAGGCACGTTGAG
CTGACCAGTGCTAAACGTCCAA
TGGCATTGCAGCTGGGGGTACAC
AGTCTTGTGTCACAGTCAGGTC
TATCTGTACTAGTCTTACGTAGT
CGGTACCCCATGACATCGACTT
GTGTCGGATCGATCATC
```

DNA code was mapped in 2001
– 3 billion letters (GATC)



How does each cell know what it is supposed to be?

```
GTAAGGCAGTTCGATATAGGTT
                                TTGAG
CTGACCAGTGCTAAACGTCCAA
TGGCATTGCAGCTGGGGGTACAC
AGTCTTGTGT

CGGTACCCCATGACATCGACTT
GTGTCGGATCGATCATC
```

Only parts of the genome are active at any one time

COL1A1 gene expressed in hair cells, but not brain cells

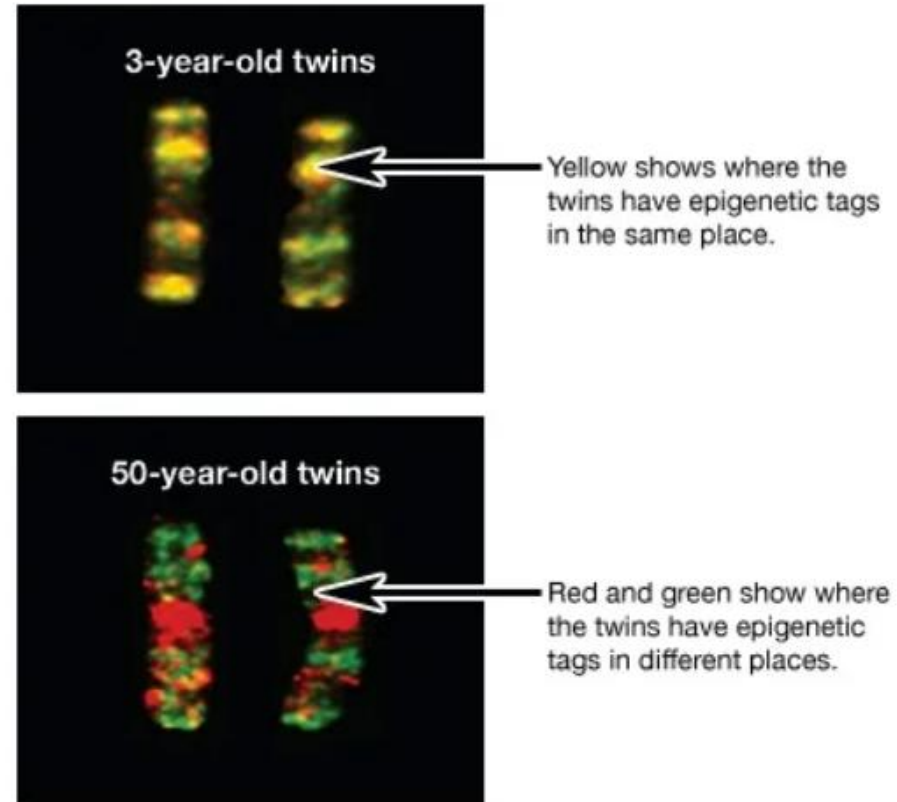
Example 1: Identical twins

Identical twins come from the same fertilized egg
→ share identical genetic profiles (genome)



→ Your behaviors and environment can cause changes that affect the way your genes work

Chromosome 3 Pairs
3-year old twins vs. 50-year-old twins



Chromosome 3 pairs in each set of twins are digitally superimposed. One twin's epigenetic tags are dyed red and the other twin's tags are dyed green. When red and green overlap, that region shows up as yellow. The 50-year old twins have more epigenetic tags in different places than do 3-year-old twins.

Example 2: Identical twins 2

Identical twins come from the same fertilized egg → share identical genetic profiles (genome)

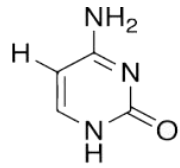


Leora Eisen

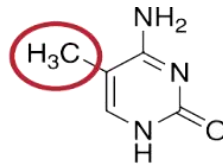
Linda Lewis
(leukemia)

→ Your behaviors and environment can cause changes that affect the way your genes work

What are DNA Methylation and genome-wide DNA methylation (Epigenome)?

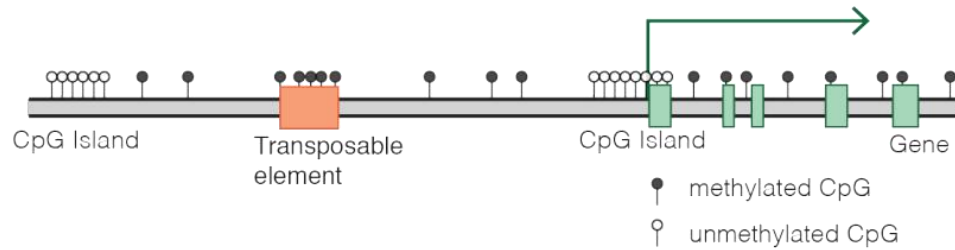
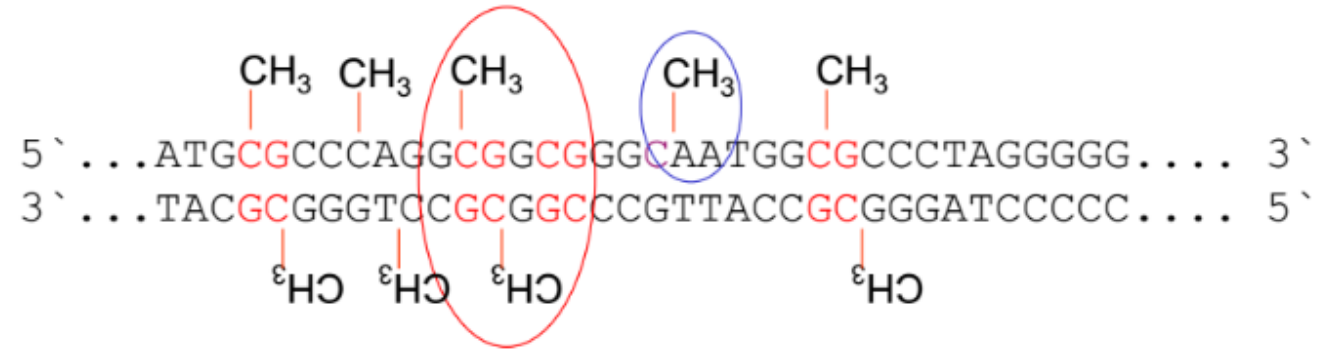


Cytosine

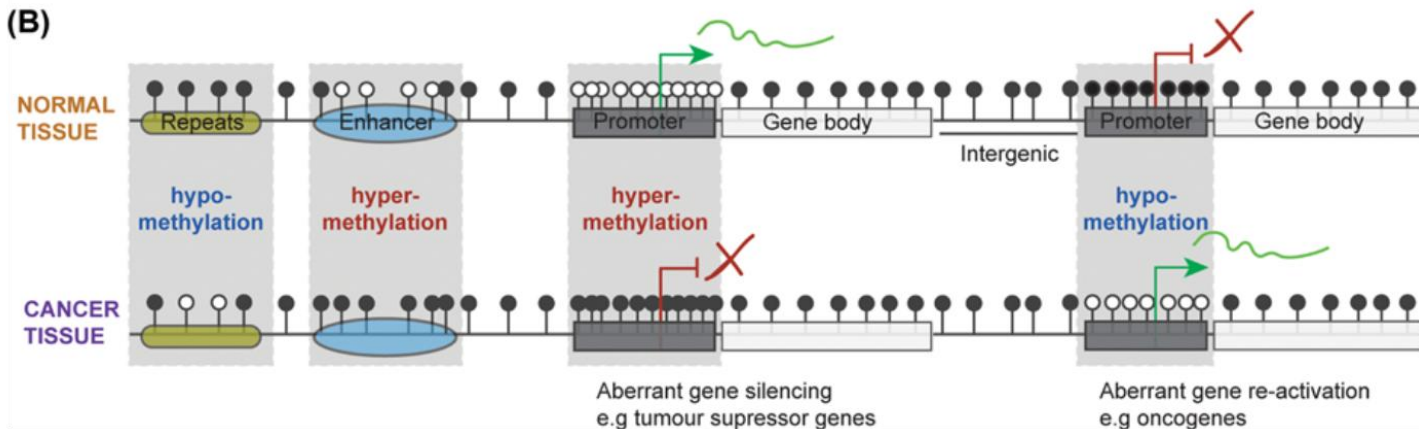


methyated Cytosine

Typical mammalian DNA methylation landscape



(B)



DNA methylation regulates gene expression

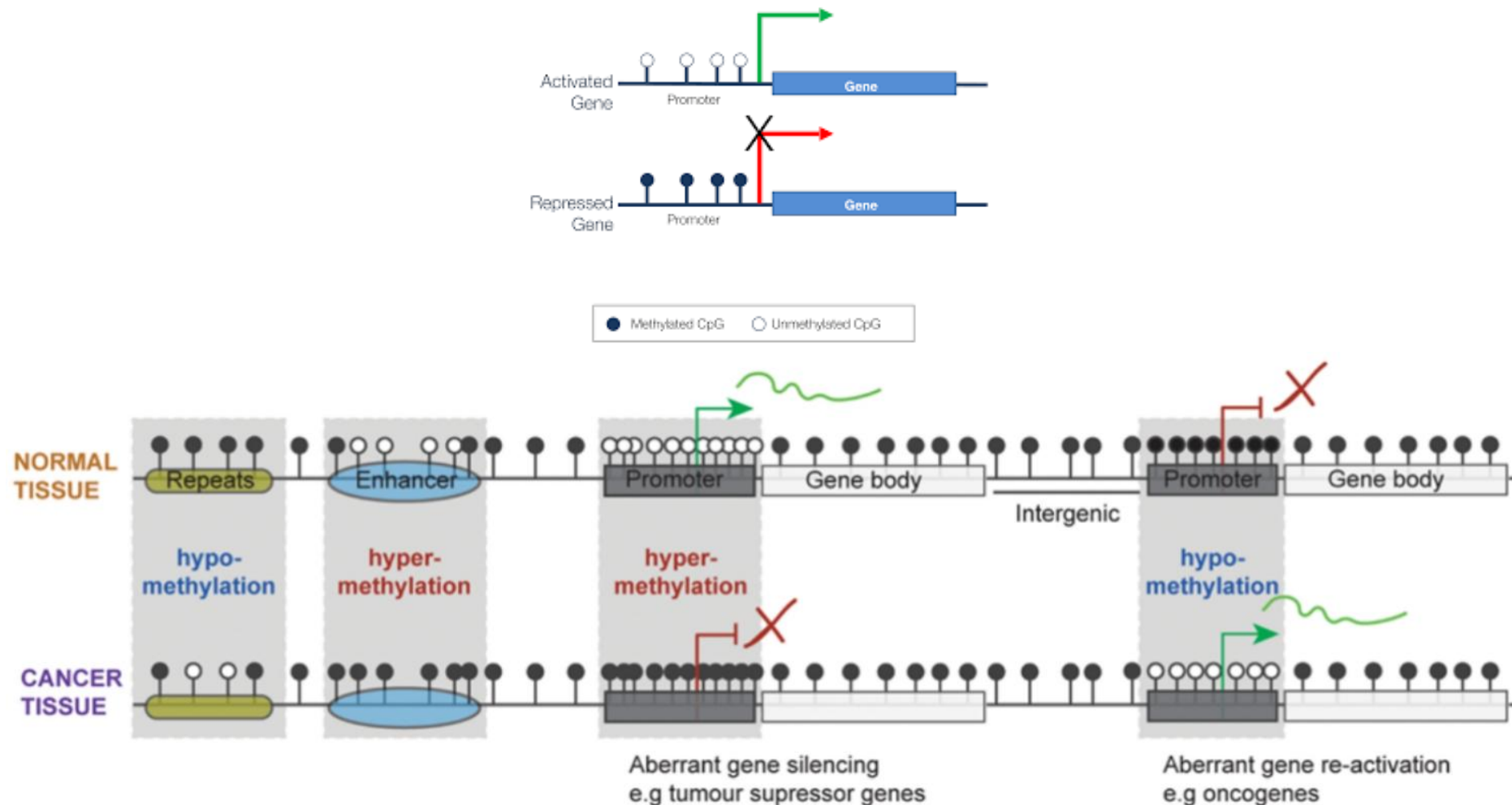
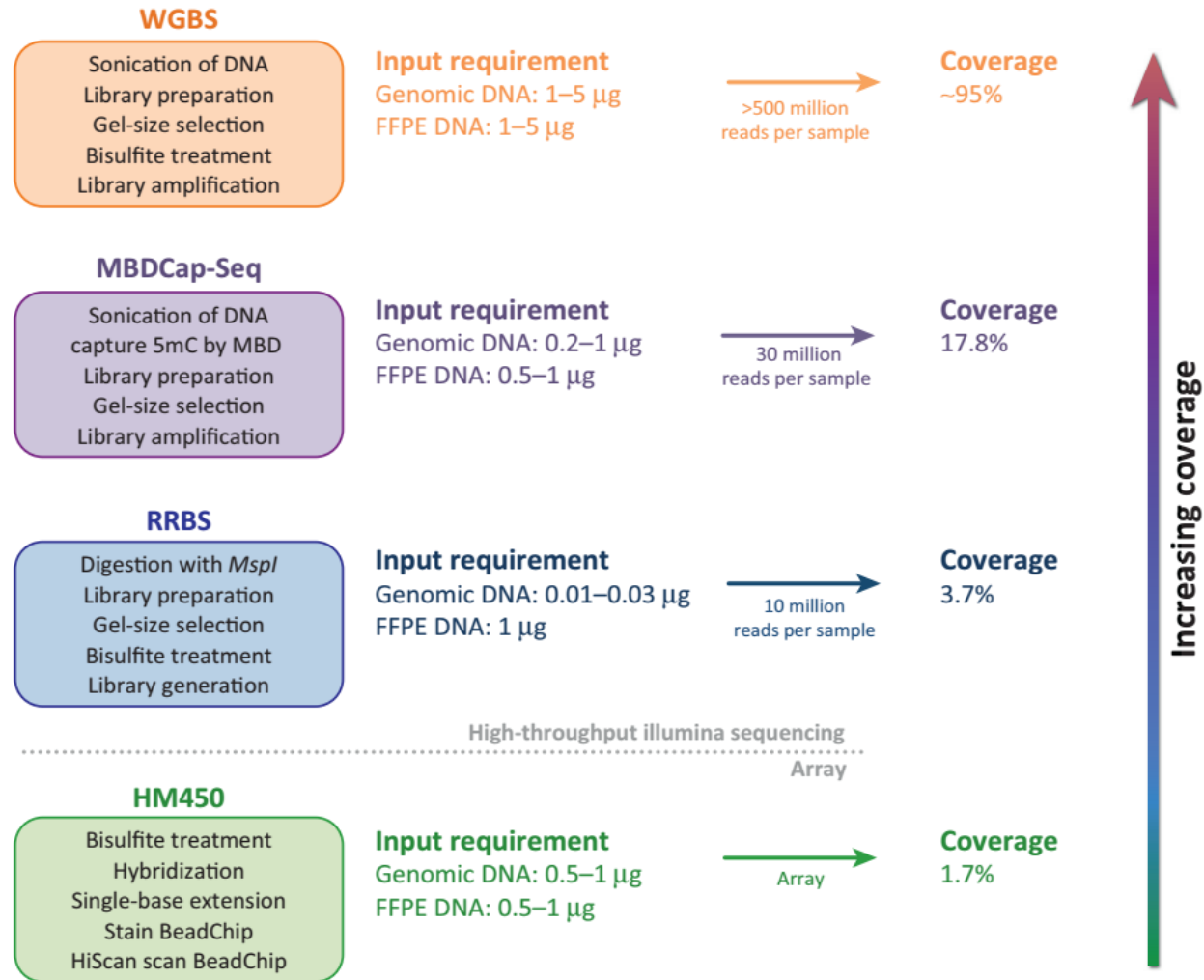


Figure 2. The aberrant gene silencing and re-activation in cancer tissue due to alteration of promoter DNA methylation. The green string is mRNA and the red cross is repression. The figure adapted from 10.1042/EBC20190037.

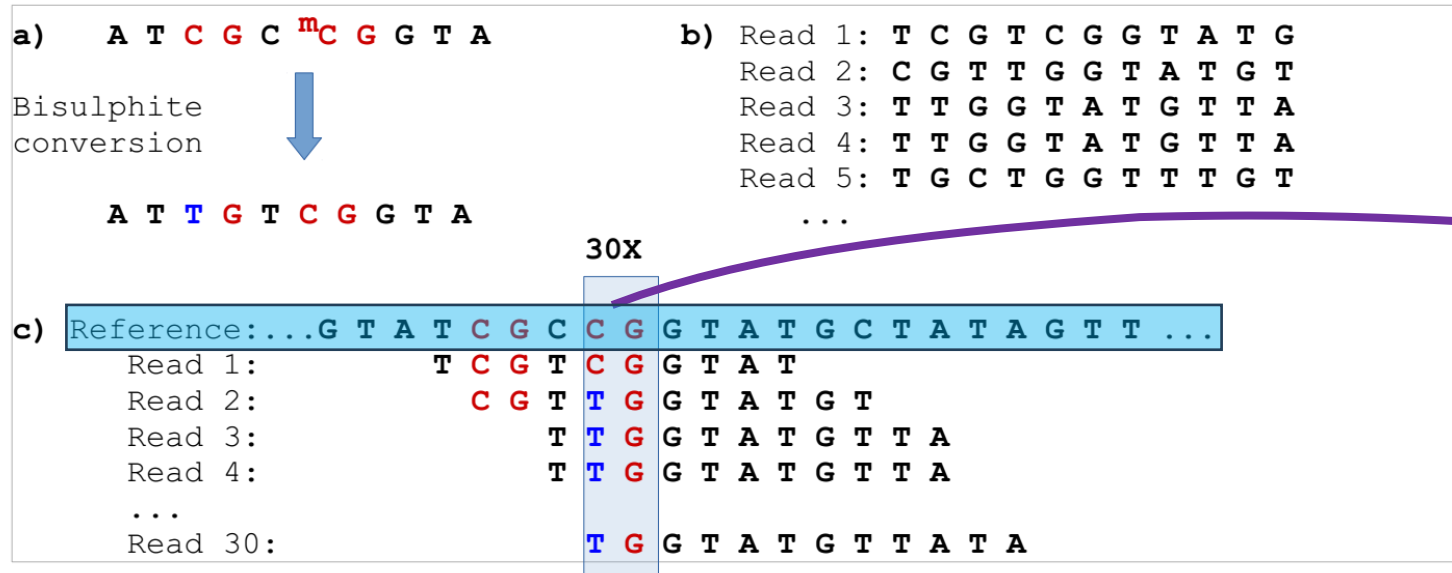
Methods to profile DNA methylation



850K/EPIC

10.1016/j.tig.2013.11.004

Bisulphite Sequencing



Methylation Beta value =
 $100 * C / (C + T) = 100 * 1 / (1 + 4)$
 = 20%

Figure 1. a) Sodium bisulphite converts unmethylated cytosines to thymines. b) The converted DNA is sequenced. c) Alignment and aggregation of these short sequence can be used to quantify the ratio of methylated (C) to unmethylated (converted T) cytosines from a population of cells at a single locus.

Total_cov_matrix_CpG.txt

	A	B	C	D
1	chr	pos	nmDNA1	nmDNA2
2	chr2	133031228	56	55
3	chr2	133031240	74	81
4	chr2	133031244	82	79
5	chr2	133031249	64	70
6	chr2	133031265	65	69
7	chr2	133031268	67	56

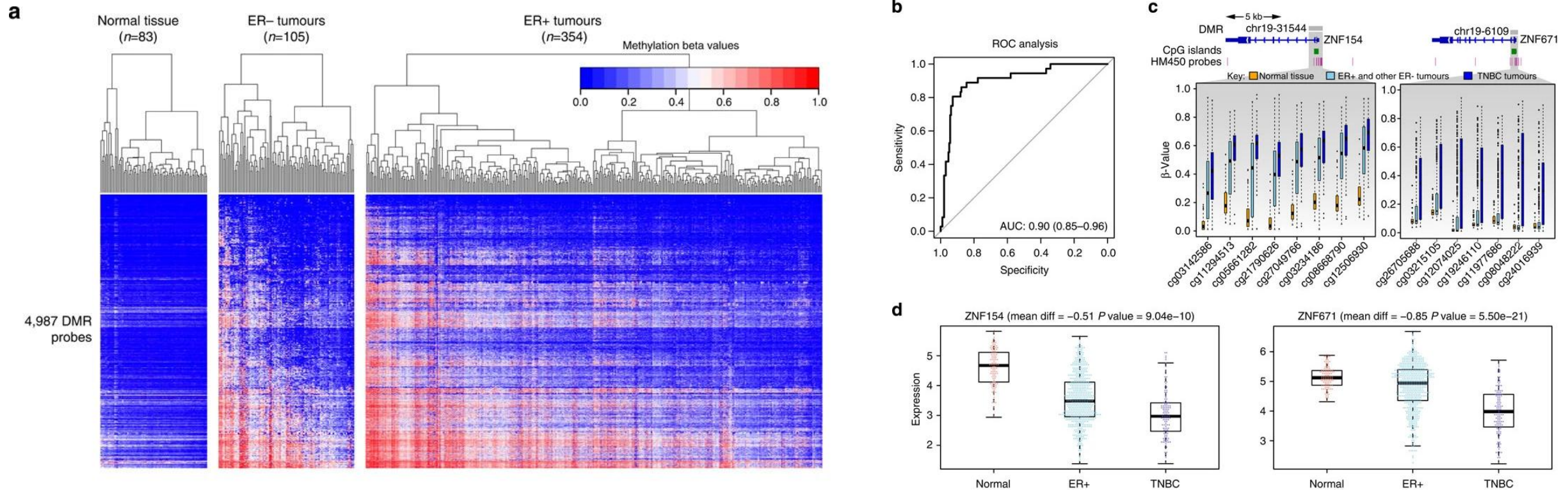
Meth_cov_matrix_CpG.txt

	A	B	C	D
1	chr	pos	nmDNA1	nmDNA2
2	chr2	133031228	41	28
3	chr2	133031240	29	19
4	chr2	133031244	47	47
5	chr2	133031249	35	24
6	chr2	133031265	19	21
7	chr2	133031268	24	16

Perc_cov_matrix_CpG.txt

	A	B	C	D
1	chr	pos	nmDNA1	nmDNA2
2	chr2	133031228	73	51
3	chr2	133031240	39	23
4	chr2	133031244	57	59
5	chr2	133031249	55	34
6	chr2	133031265	29	30
7	chr2	133031268	36	29

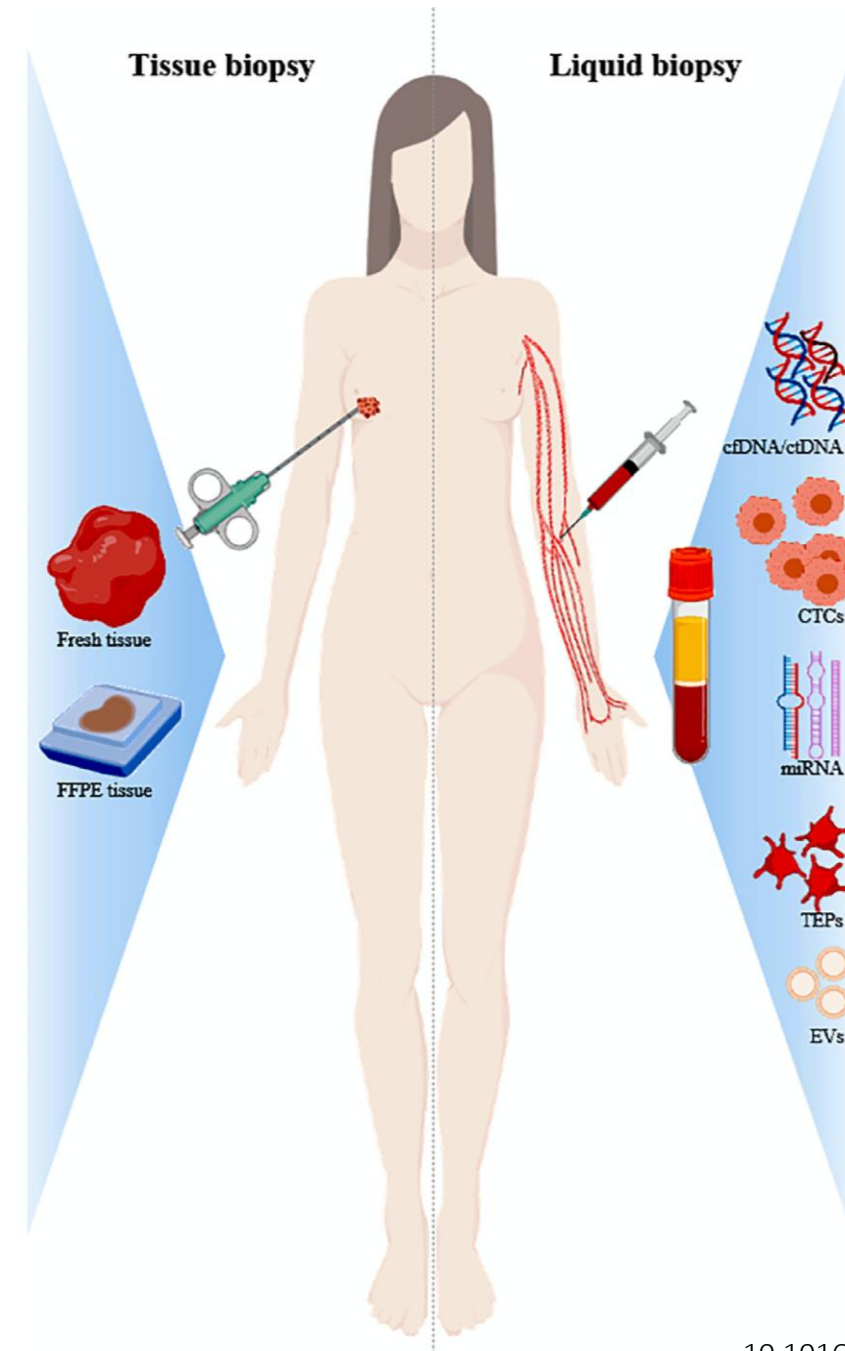
DNA methylation are promising biomarkers for screening and early detection of breast cancer



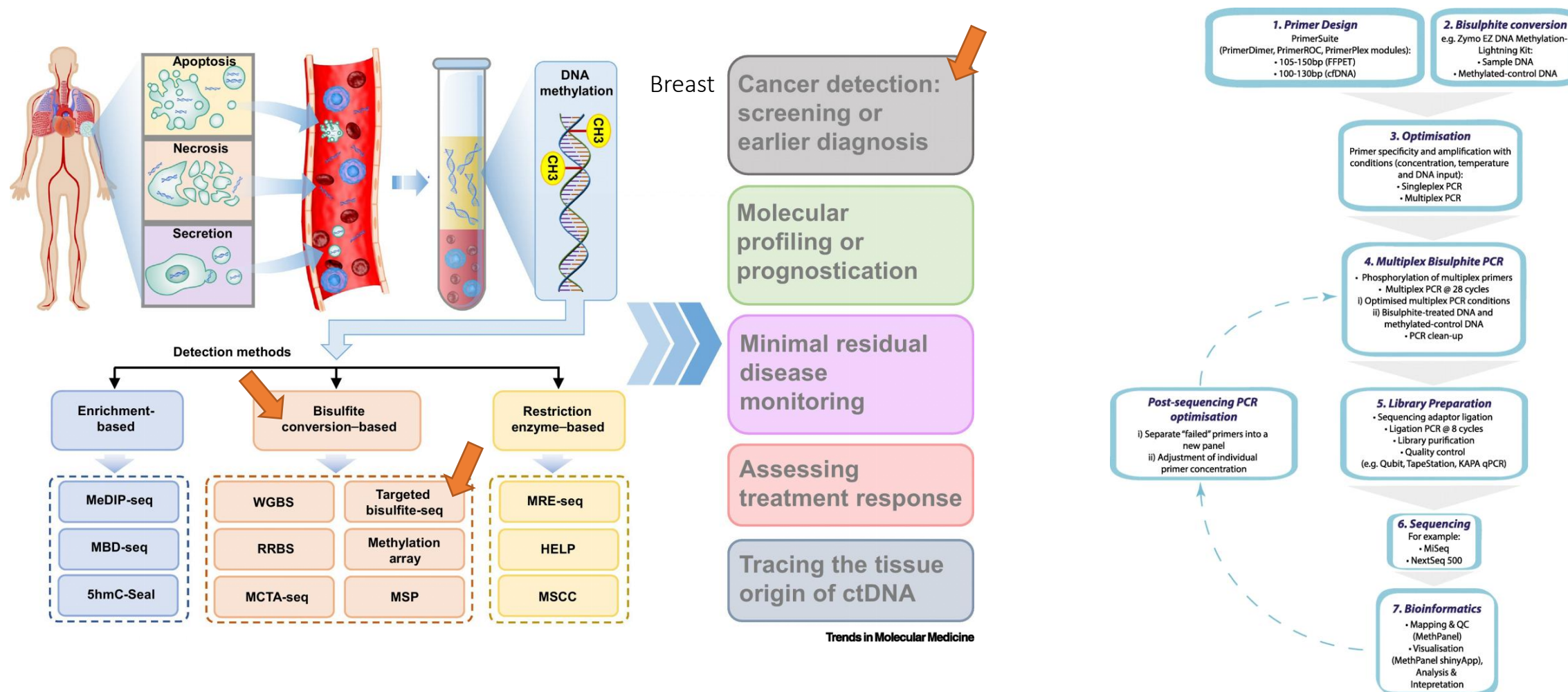
Stirzaker C., Zotenko E., Song J.Z., Qu W., Nair S.S., Locke W.J., Stone A., Armstrong N.J., Robinson M.D., Dobrovic A... **SJ. Clark**. Methylome sequencing in triple-negative breast cancer reveals distinct methylation clusters with prognostic value. Nat. Commun. 2015

Aims of the study

- Build up DNA methylation panel for early detection of breast cancer
- Validate the panel
 - Tissue biopsy of patients and healthy controls (**Tissue/FFPE**)
 - Liquid biopsy of patients and healthy controls (**cfDNA**)



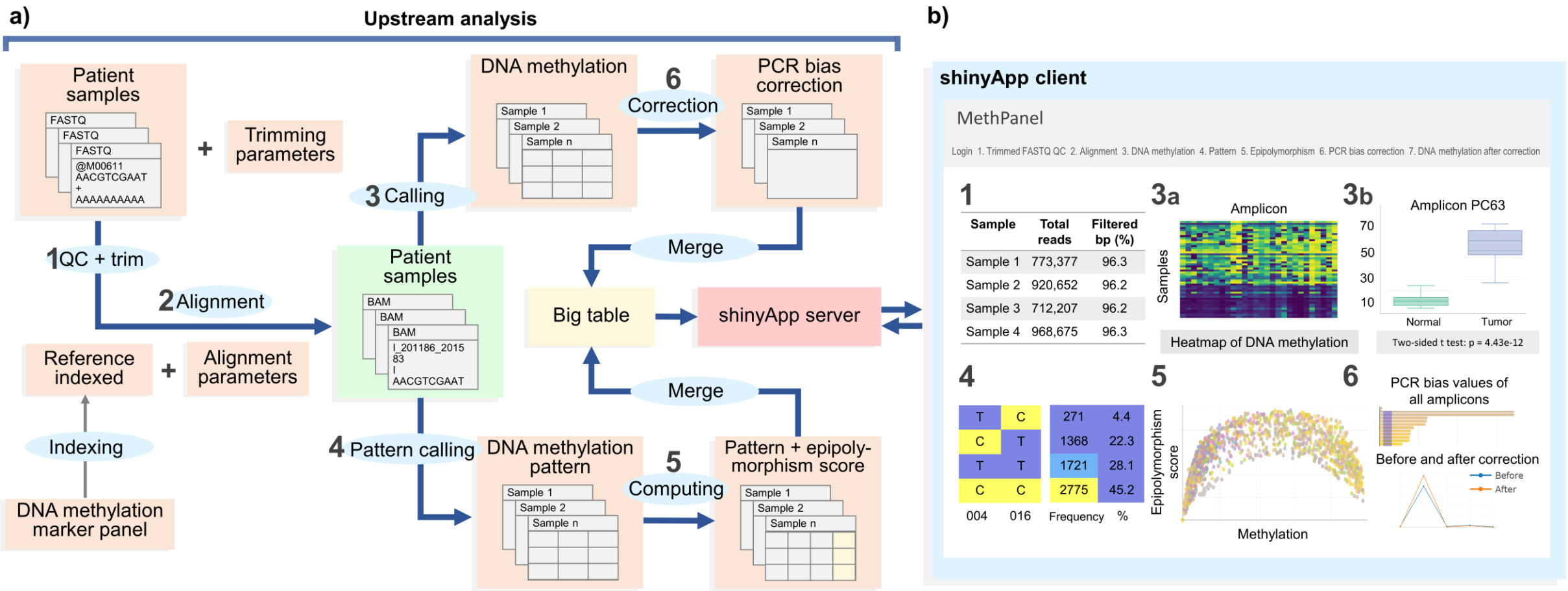
Targeted multiplex bisulphite PCR sequencing



Lam D*, Luu PL*, Song JZ, Qu W, Risbridger GP, Lawrence MG...SJ Clark, R Pidsley, Stirzaker C. Comprehensive evaluation of targeted multiplex bisulphite PCR sequencing for validation of DNA methylation biomarker panels. Clin Epigenetics. 2020;12(1):90. * Equal contribution authors

Luo H., Wei W., Ye Z., Zheng J., Xu R. Liquid biopsy of methylation biomarkers in cell-free DNA. Trends Mol. Med. 2021

MethPanel: analysing and visualizing amplicon bisulphite sequencing data

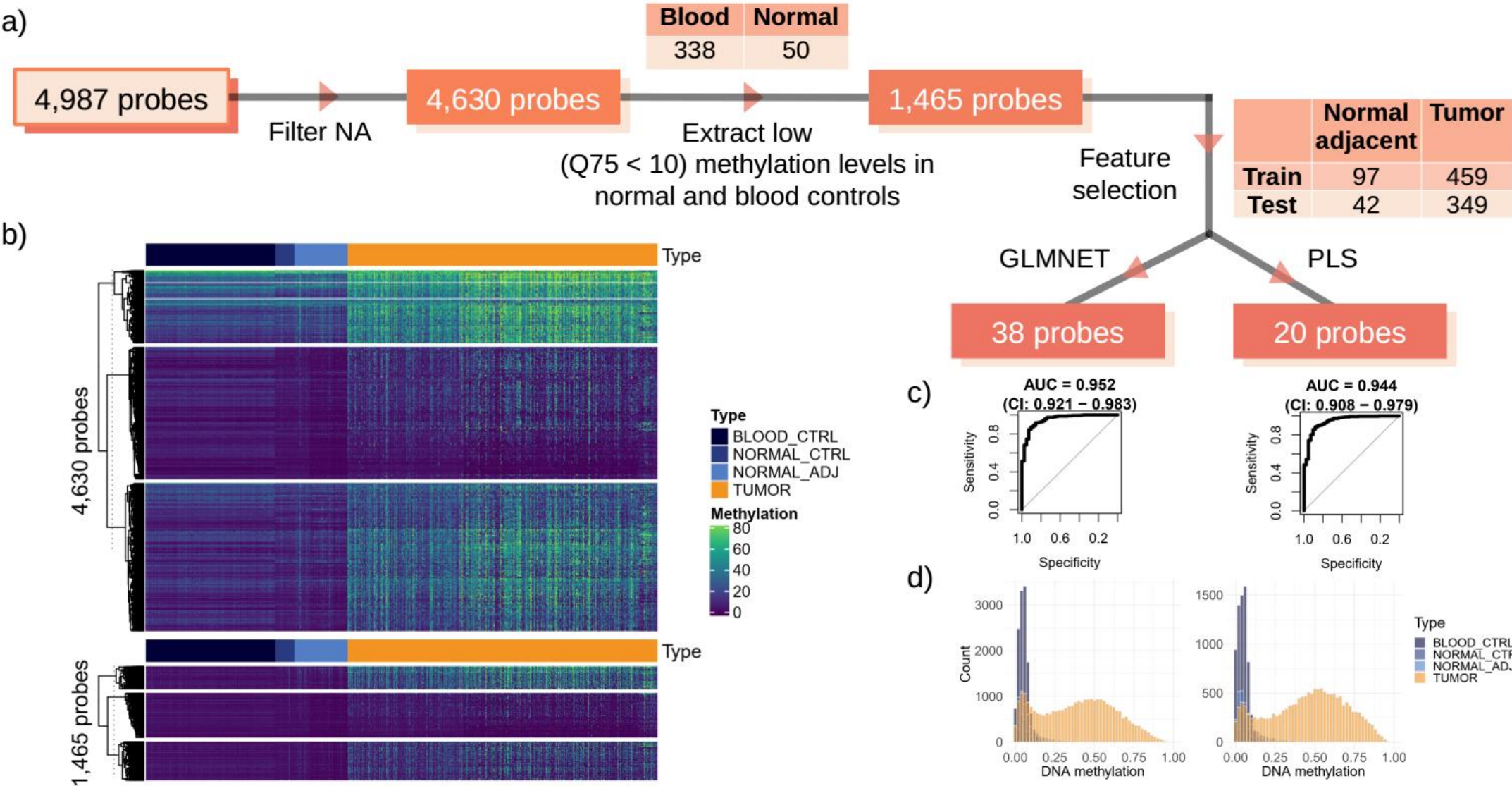


PL Luu, PT Ong, TTH Loc, D Lam, R Pidsley, C Stirzaker, SJ Clark, **MethPanel**: a parallel pipeline and interactive analysis tool for multiplex bisulphite PCR sequencing to assess DNA methylation biomarker panels for disease detection, *Bioinformatics* (2020).

Machine learning techniques

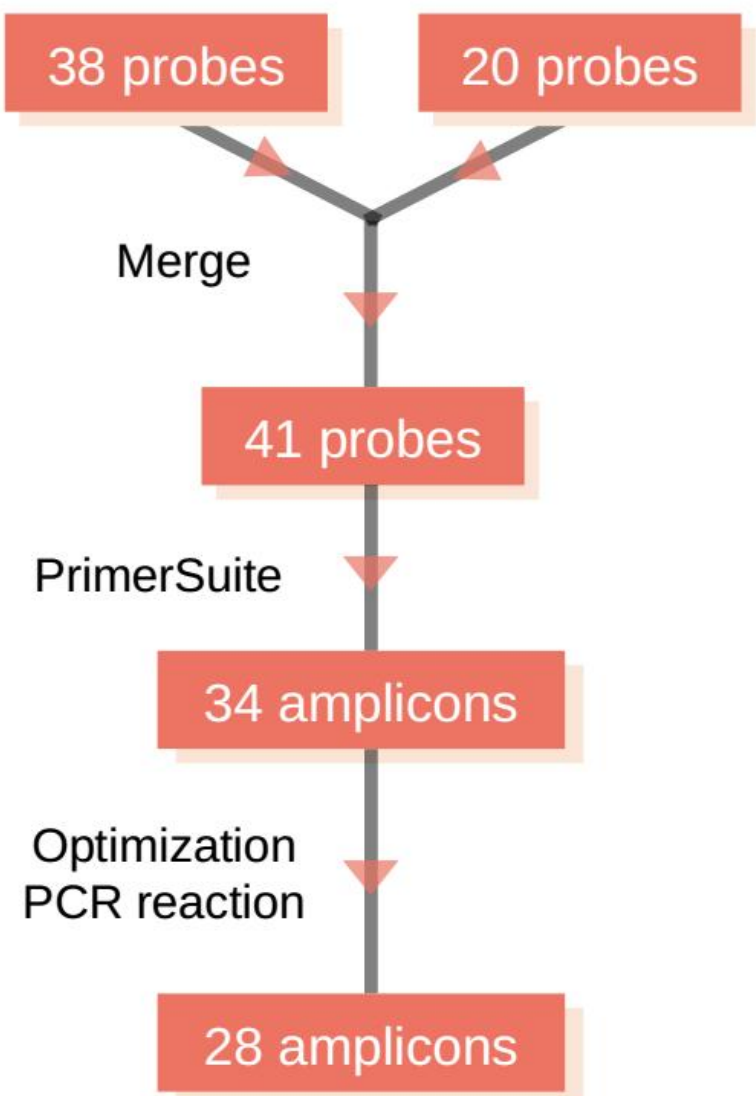
- Gmlnet, PLS and MVUR for feature selection
- Random Forest for classification
- Smote for up sampling in unbalance dataset of cfDNA

Build up DNA methylation breast cancer panel

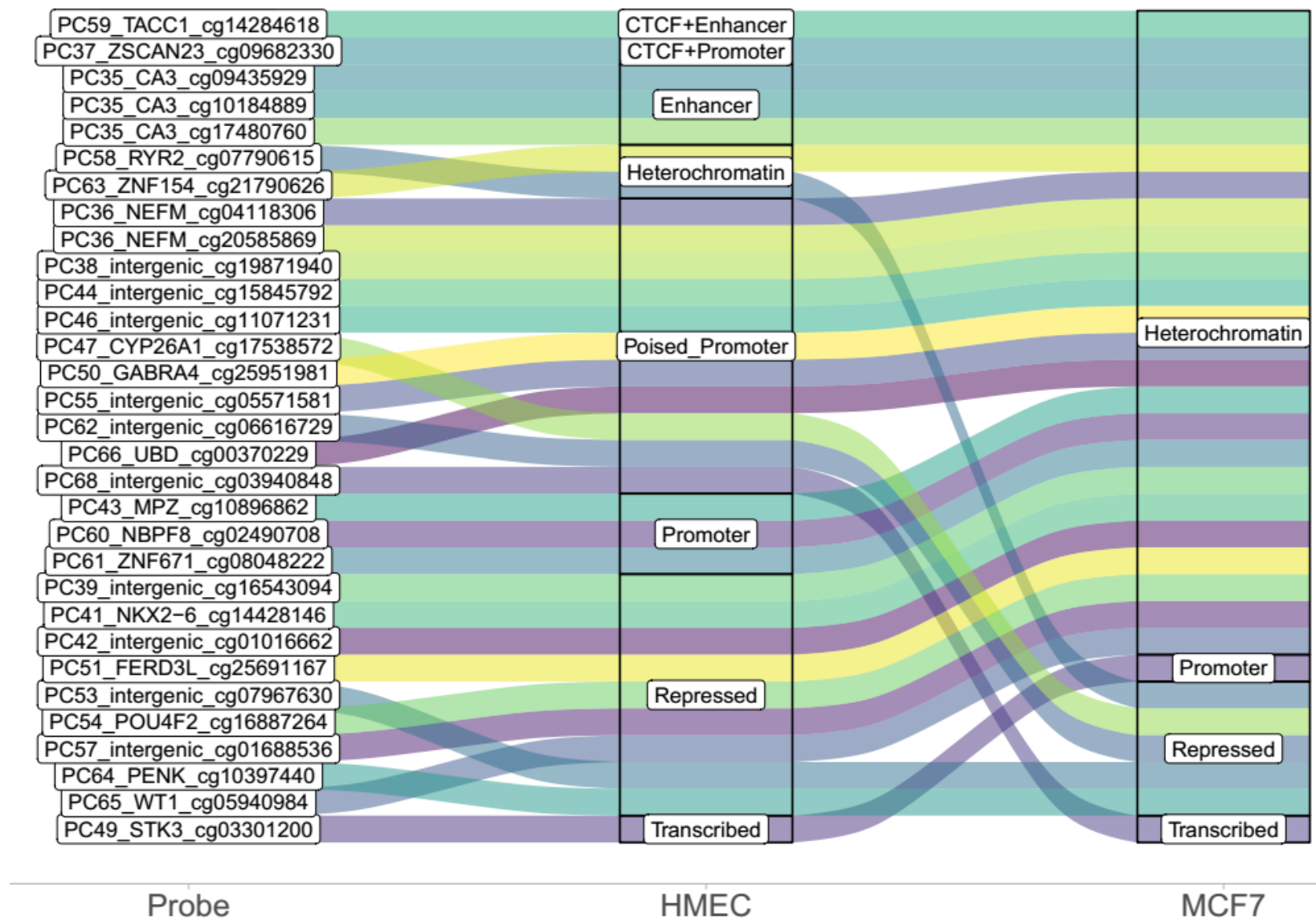


Characterization of DNA methylation breast cancer panel

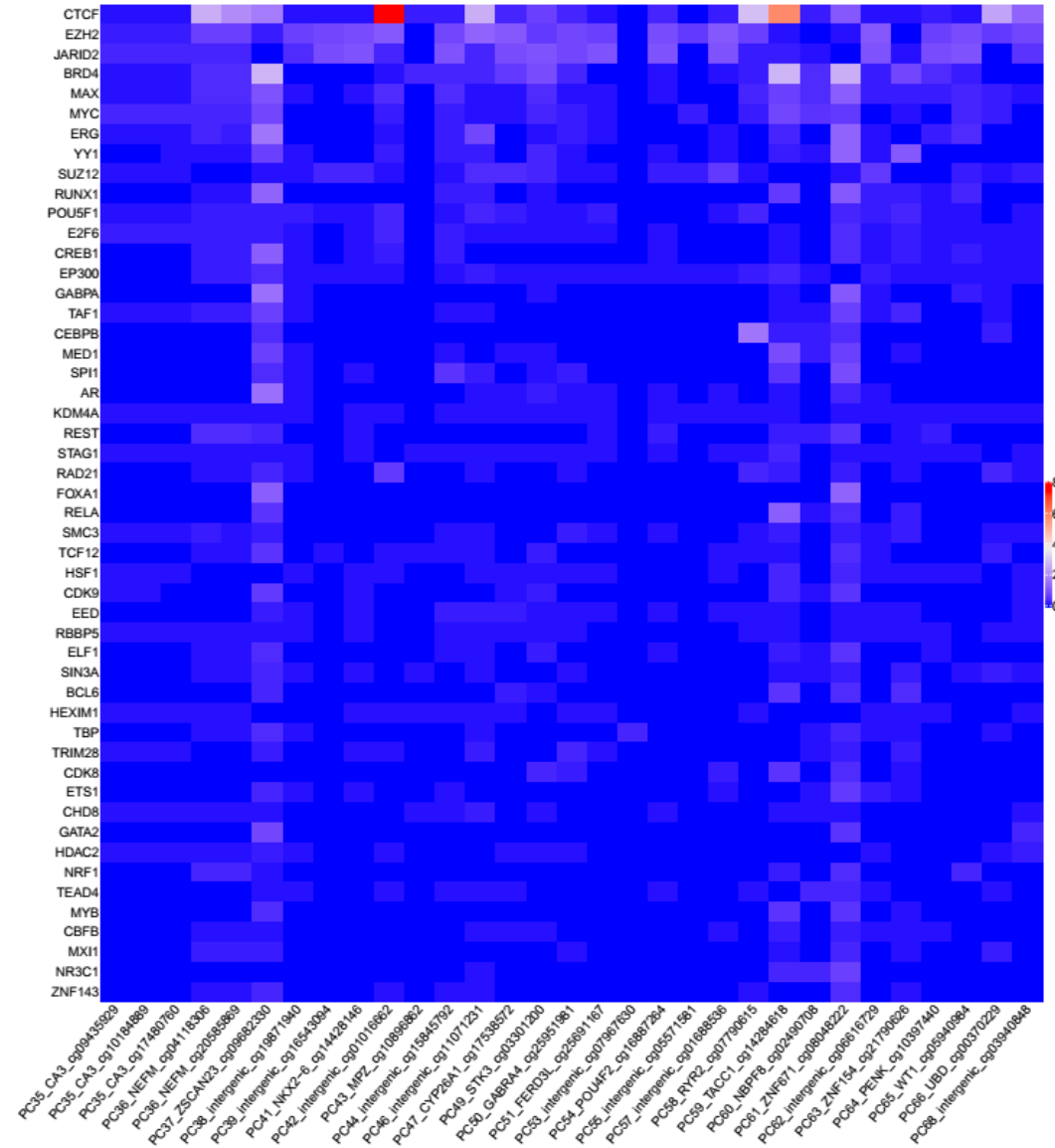
a)



b) ChromHMM

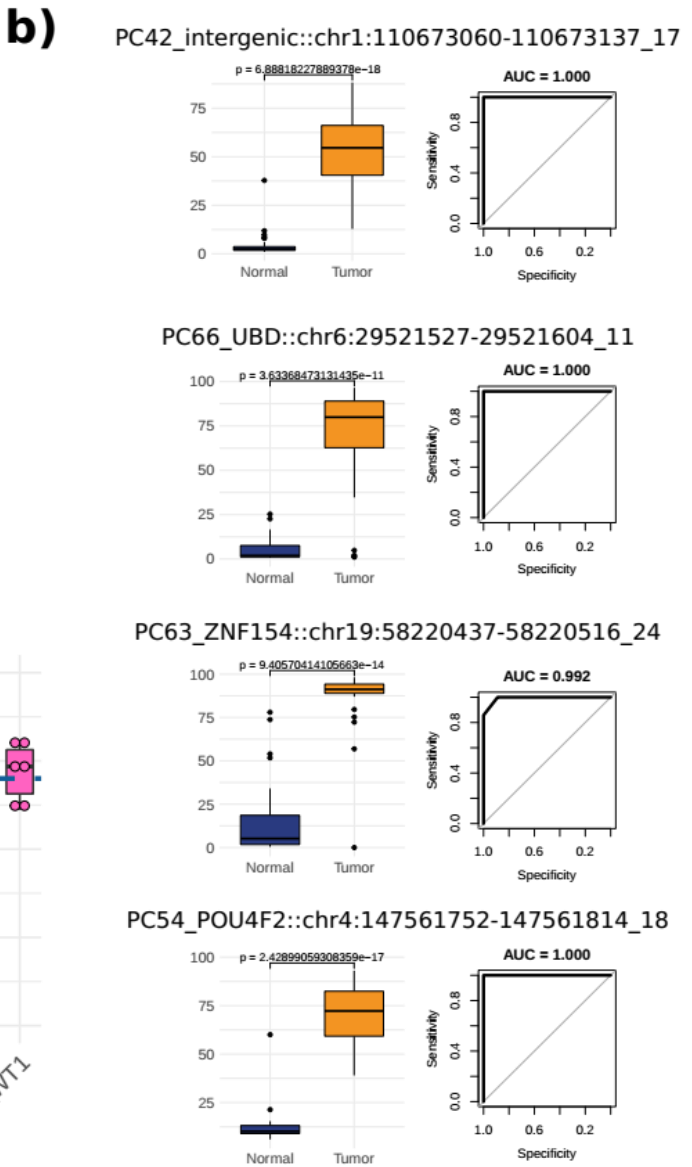
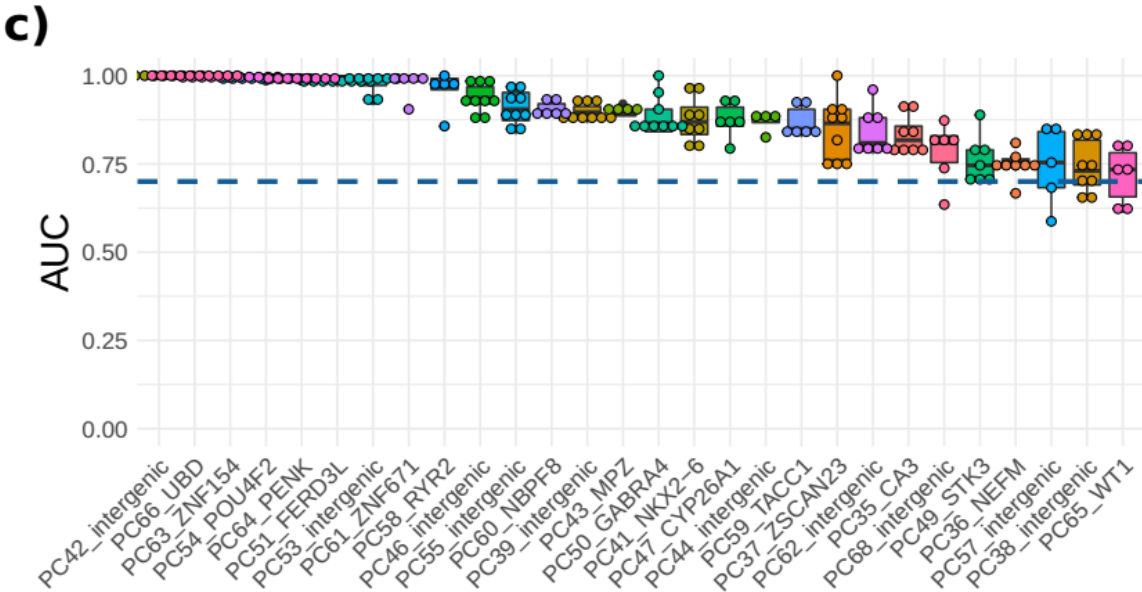
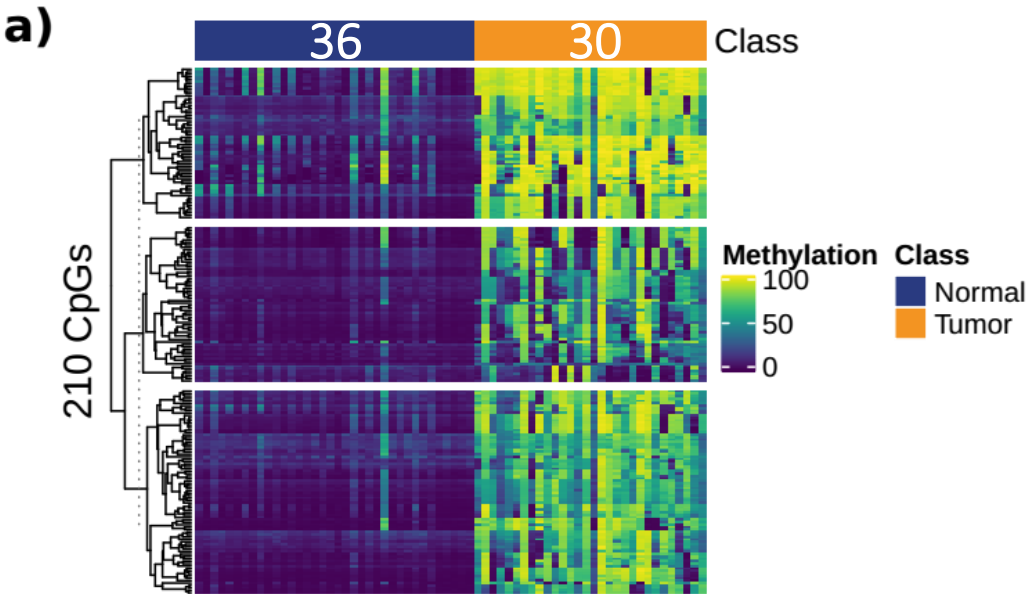


c) TFBS



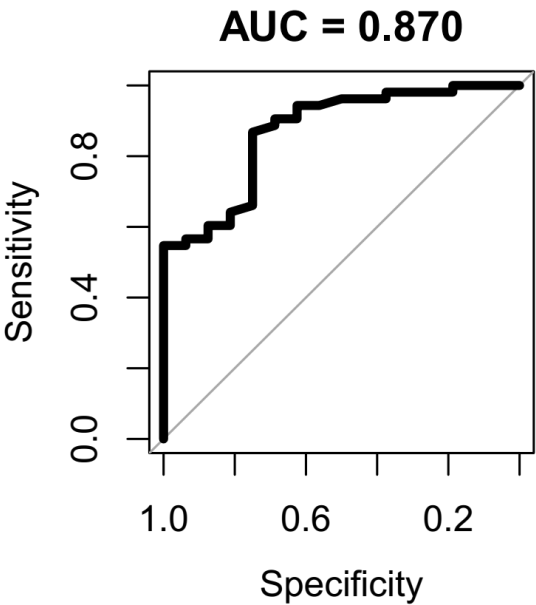
Validation of the breast cancer panel on Tissue/FFPE

Normal (36)		Tumor (30)
Blood	Normal	Tumor
20	16	30

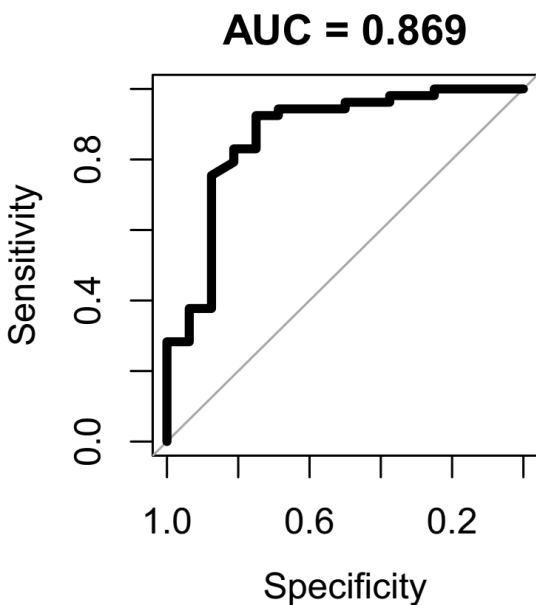


Validation of the breast cancer panel cfDNA

Liquid Biopsy (cfDNA)		
Normal (66)		Tumor (215)
Blood	Normal	Tumor
20	46	215

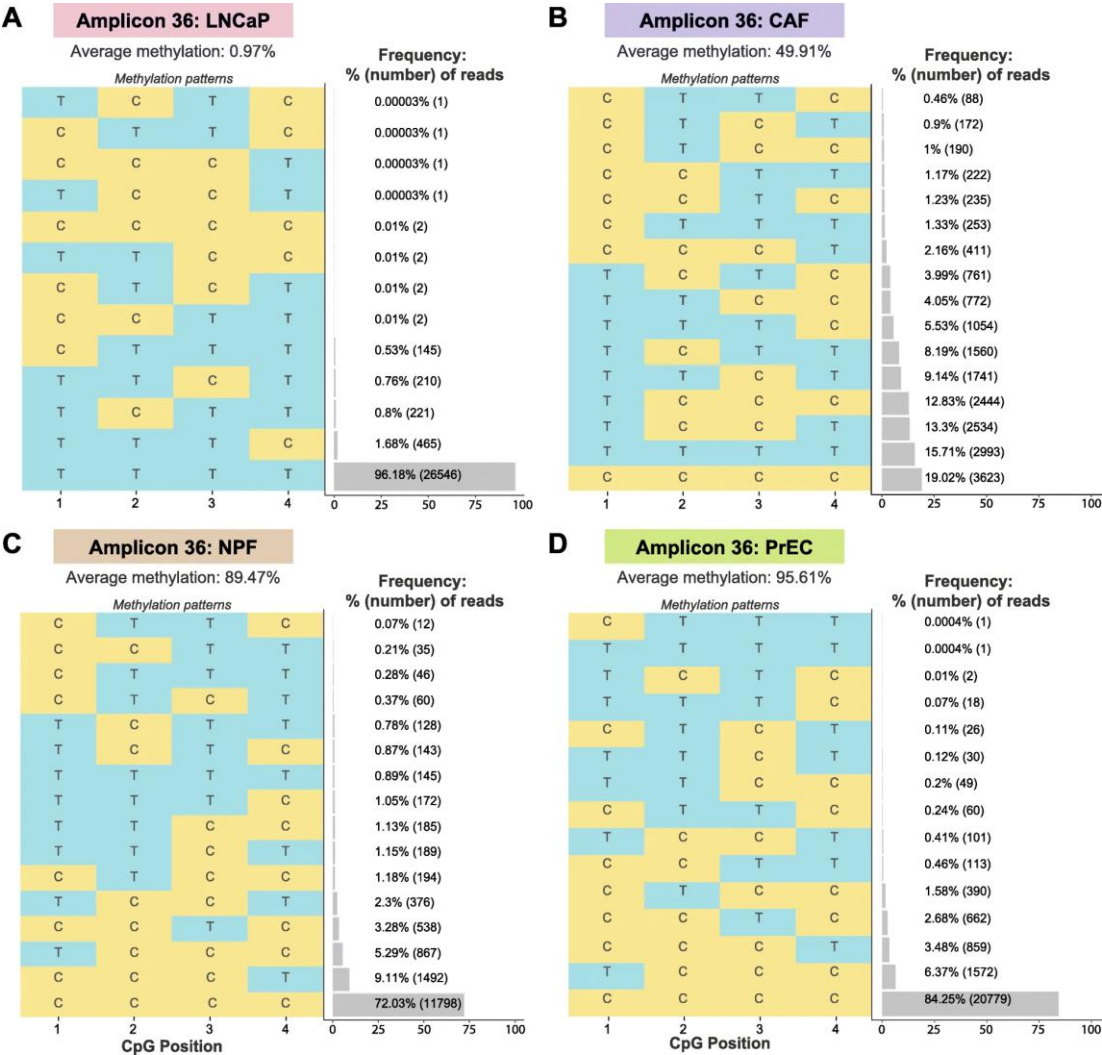


10 significant CpGs (T test, p value < 0.05)

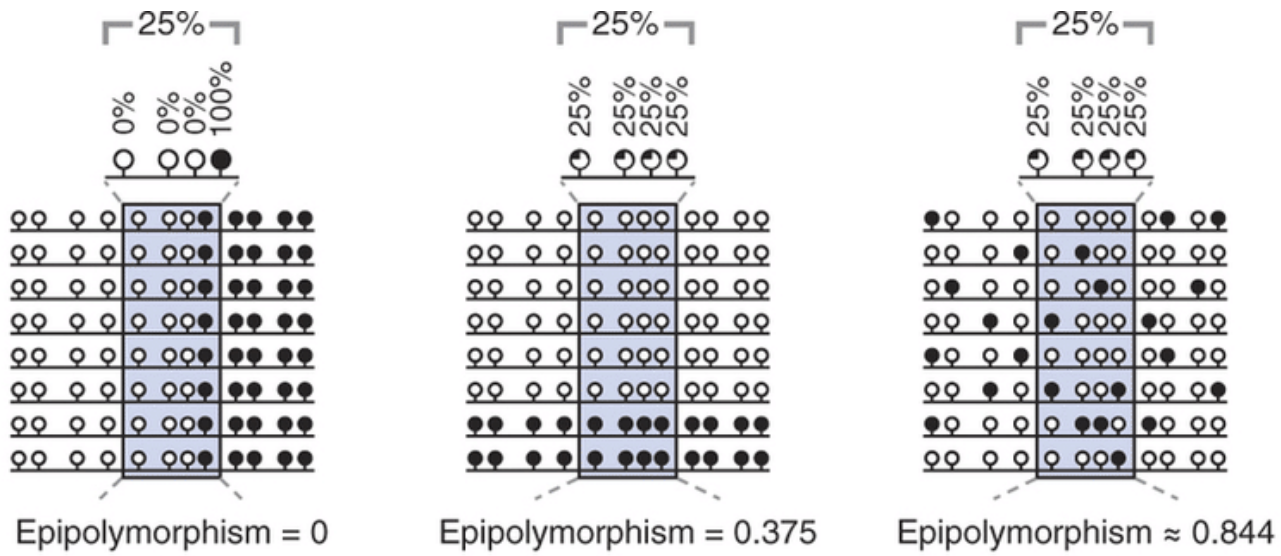


6 important CpGs (from feature selection package MVUR)

DNA methylation pattern

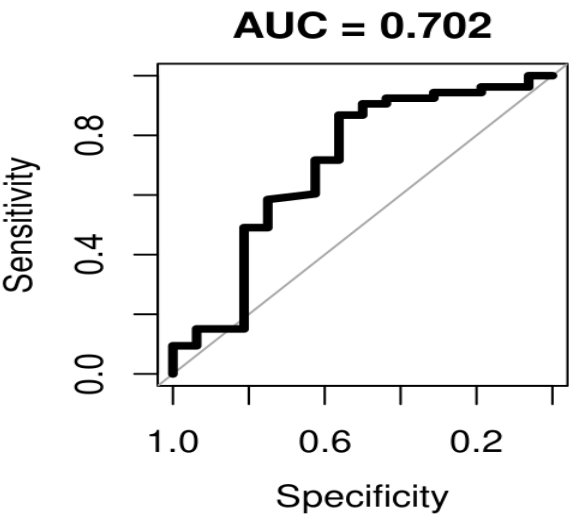


Epipolymorphism

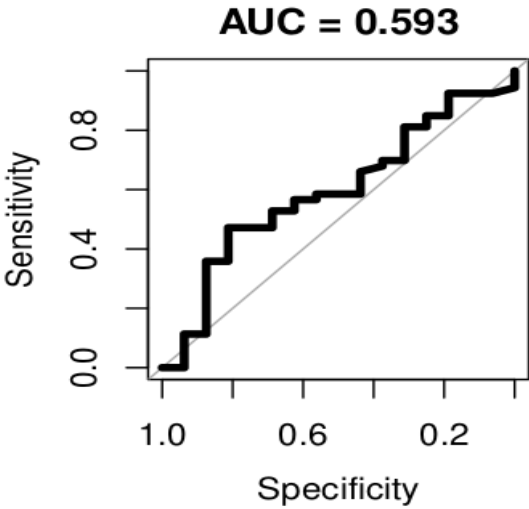


Pattern and EpiPolymorphism score on cfDNA

Liquid Biopsy (cfDNA)		
Normal (66)		Tumor (215)
Blood	Normal	Tumor
20	46	215



Pattern



EpiPolymorphism score

EpiClass

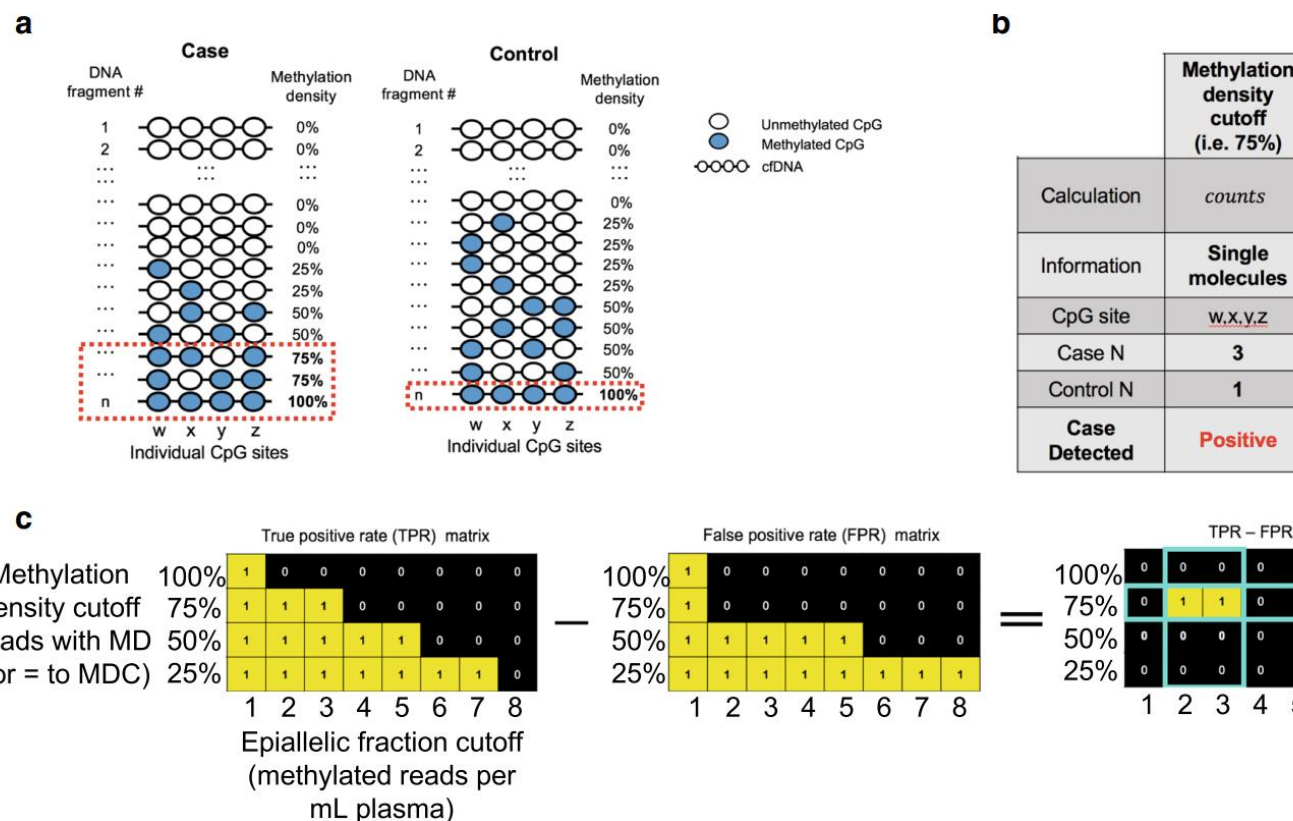
METHODOLOGY

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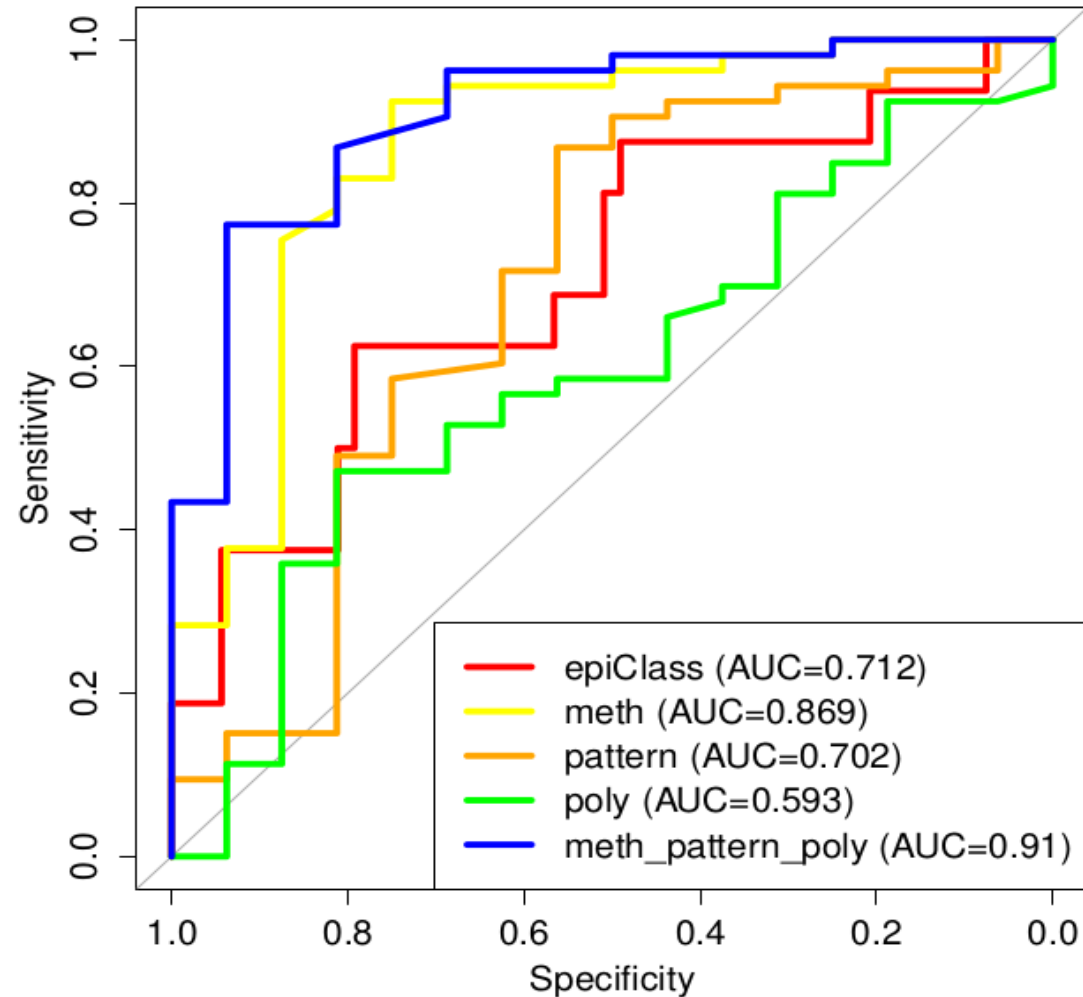


Leveraging locus-specific epigenetic heterogeneity to improve the performance of blood-based DNA methylation biomarkers

Brendan F. Miller¹, Thomas R. Pisanic II^{2*}, Gennady Margolin¹, Hanna M. Petrykowska¹, Pornpat Athamanolap⁵, Alexander Goncarencu¹, Akosua Osei-Tutu³, Christina M. Annunziata³, Tza-Huei Wang^{2,4,5} and Laura Elnitski^{1*}

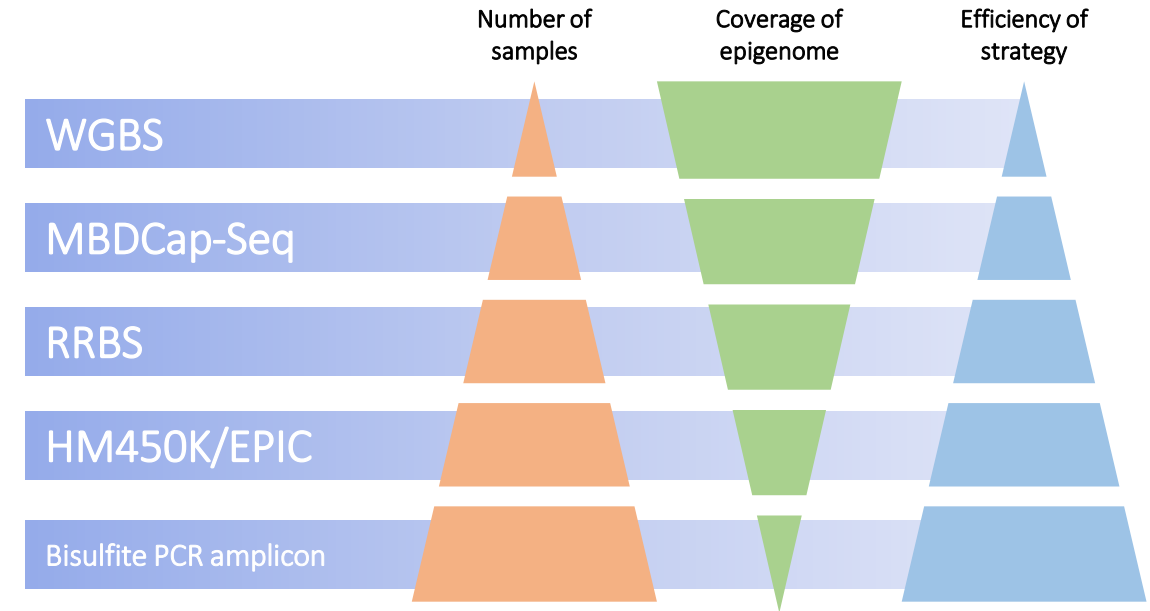


The power of combination of DNA methylation, Pattern and EpiPolymorphsim score for diagnosis of breast cancer using cfDNA (AUC=0.91)



Conclusion

- We succeed to build up the breast cancer DNA methylation panel.
- The breast cancer panel of DNA methylation biomarkers is feasible for diagnosis of both tissue and liquid biopsy.
- The combine DNA methylation status of different CpGs or with the other features such as methylated patterns and epi-polymorphism showed better performance (AUC = 0.91) than a single one.

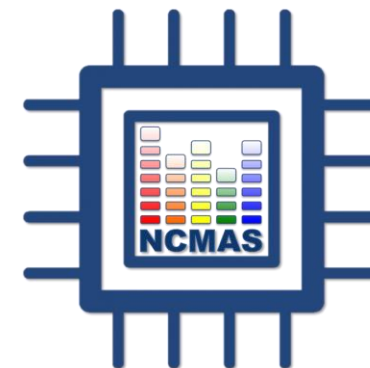


The best performance amplicons in the panel

Acknowledgement

- Prof. Susan Clark
- Dr Clare
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- Dr Truong
- Dr Ngu
- Dr Duy Thanh
- Dr Phuong
- Dr Ruth
- Jenny
- Wenija
- Dilys
- Dr Ong
- Thong
- Dr Elena
- Dr Qian
- Dr Amanda
- Dr Jo
- Dr Braydon
- Dr Shalima

All patients who donate samples



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Xin chân thành cảm ơn!

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